Understanding Multilateral Tariffs and Exchange Rates*

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Abstract

The textbook prediction that an import tariff appreciates the tariffing country's currency comes from a two-country world, in which the bilateral exchange rate, the effective exchange rate and the terms of trade are the same object. With many countries they come apart. We characterize the long-run exchange-rate response to arbitrary tariff vectors in an N-country trade model. To first order, the response is the product of the tariff's trade-balance disturbance at unchanged exchange rates and the inverse of the exchange-rate adjustment matrix, and we state a multilateral Marshall–Lerner condition under which that adjustment is well behaved. Under partner symmetry, any tariff vector decomposes into *average protection* and *relative treatment*: average protection alone moves the nominal effective exchange rate, and relative treatment alone moves bilateral rates around it.

In a nested-CES model these responses are governed by two margins: diversion toward the domestic economy, ρ_D, and substitution across foreign suppliers, ρ. A unilateral tariff always appreciates the tariffing country's effective exchange rate, yet it depreciates the currency against every untariffed bystander when $\rho > N\rho_D$, which requires that the two elasticities differ. Under retaliation the threshold falls to ρ_D, and both belligerents depreciate in effective terms. The symmetric N-country economy reduces exactly to a three-country economy with a composite bystander, which makes the role of N transparent. Under partner asymmetry these clean conclusions can fail. Most notably, a tariff on a small, low-weight partner can worsen the tariffing country's aggregate terms of trade, and hence its welfare, when foreign suppliers are close substitutes. Read stylistically, the 2025 episode fits this logic: retaliation and high substitutability between China and other suppliers both point toward a weaker dollar against bystanders, while the trade channel cannot account for dollar weakness once tariffs became close to uniform.

Keywords: tariffs, exchange rates, trade diversion, terms of trade, multilateral Marshall–Lerner condition.
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1 Introduction

What happens to exchange rates when a country imposes tariffs? The conventional answer, going back to Lerner (1936), Metzler (1949), Mundell (1961) and Krugman (1982), is that the tariffing country's currency appreciates. A tariff shifts expenditure away from imports and improves the trade balance at the prevailing exchange rate. Under the Marshall–Lerner condition, restoring external balance then requires an appreciation. The prediction is robust *within* the two-country model: it follows from monotonicity properties of import and export demand rather than from functional forms.

The 2025 tariff episode is hard to square with this prediction. After the April 2 announcements the dollar weakened, notably against the euro, while moving little against the renminbi. Much of the recent literature has turned to financial explanations. These include tariff uncertainty and currency risk premia (Kalemli-Özcan et al., 2026), the erosion of the dollar's convenience yield and hedging value (Acharya and Laarits, 2026; Jiang et al., 2025), and valuation effects on dollar liabilities (Itskhoki and Mukhin, 2025). Ostry et al. (2025) document a striking regularity across the 2018–20 and 2025 episodes: the dollar appreciates when US tariffs are unilateral and depreciates when partners retaliate, and in US–China tit-for-tat episodes it is roughly stable against the renminbi. They interpret this pattern through dominant-currency pricing.

This paper asks a prior question: **what does the real trade channel itself imply for exchange rates once trade is multilateral?** The two-country model cannot answer it, because it has only one exchange rate, and substitution away from a tariffed supplier can only go toward domestic goods. With more than two countries, three things that coincide when $N = 2$ come apart: the adjustment of trade balances to exchange rates, the allocation of tariff-induced expenditure switching across foreign suppliers, and the distinction between bilateral and effective exchange rates. Our main message is that **the conventional wisdom survives multilateralization as a statement about the effective exchange rate but fails as a statement about bilateral exchange rates.** The model characterizes precisely the substitution, network and policy conditions under which each holds.

We work in a static, long-run, real environment: one factor, linear technologies, balanced trade and no nominal rigidities. In this class of models the nominal exchange rate equals the relative wage in a common currency, and therefore the bilateral terms of trade. This is deliberate. It isolates what trade flows alone pin down, and it lets us obtain sharp analytical results. Our results should be read as the long-run exchange-rate position that the trade channel requires once prices have adjusted, conditional on the tariffs persisting.

Results. We proceed in three steps.

General framework (Section 2). In any differentiable N-country trade environment, the first-order exchange-rate response to a tariff change is $d\nu = (-J)^{-1}F\,ds$. Here F is the vector of trade-balance disturbances generated by the tariff at unchanged exchange rates, and J is the Jacobian of trade balances with respect to exchange rates. We define a *multilateral Marshall–Lerner* (MML) condition: $-J$ is a nonsingular M-matrix. It is implied by gross substitutability of trade balances in currency values, and it is the natural heir of the Hicks–Metzler–Mosak stability conditions. MML ensures that exchange-rate adjustment is stabilizing. It does not determine the sign of any bilateral response, because F is now a vector whose entries can have different signs.

The symmetric case (Section 3). When the tariffing country's partners are symmetric, any tariff vector splits into *average protection*, the common component, and *relative treatment*, the zero-sum component. These are the eigen-directions of both J and the tariff-incidence matrix. Average protection alone determines the nominal effective exchange rate (NEER); relative treatment alone determines the dispersion of bilateral rates around it. In a nested-CES model with a home–import elasticity η and a cross-origin elasticity ρ, three margins govern everything:

- the *domestic diversion margin* $\rho_D = \alpha_D\eta + (1-\alpha_D)(1-\alpha_T)$, which combines substitution toward domestic tradables with the income effect of rebated tariff revenue;
- the cross-origin elasticity ρ;
- a *multilateral adjustment margin* M_N.

A unilateral tariff on one partner always appreciates the NEER and always appreciates the currency against the target. It depreciates the currency against each untariffed bystander if and only if $\rho > N\rho_D$. With realistic home bias this requires $\rho > \eta$, so the reversal is impossible in any single-elasticity specification. Bilateral retaliation lowers the bystander threshold from $N\rho_D$ to ρ_D and leaves the belligerents' bilateral rate unchanged; both belligerents then depreciate against every bystander and in effective terms. This is exactly the configuration-dependence that Ostry et al. (2025) document, here produced with flexible prices and no financial channel.

Finally, we show that the symmetric N-country economy reduces *exactly* to a three-country economy: the tariffing country, the target, and a composite bystander with weight $(N-2)/(N-1)$ in the tariffing country's imports and an effective home–import elasticity that blends η and ρ. This three-country representation is our workhorse for intuition. It shows that N raises the bystander threshold not because the bystander bloc is larger, but because the bloc's trade is increasingly internal and highly substitutable. Bystander size

alone moves the threshold in the opposite direction.

Asymmetry (Section 4). Away from symmetry the decomposition no longer separates cleanly, and some intuitive conclusions of the symmetric case can fail. We focus on the aggregate terms of trade. At free trade it equals the first-order welfare effect of a tariff, and under symmetry a unilateral tariff always improves it. With asymmetric partners it can deteriorate. A tariff on a partner that is small, or carries little weight in the tariffing country's imports, can produce a bystander depreciation that is large and heavily weighted enough to outweigh the gain against the target when ρ is high. The trade-weighted NEER can likewise depreciate, and the relationship with ρ need not even be monotone. We then revisit the 2025 episode in a deliberately stylized way, without calibrating.

What we do not do. The model abstracts from nominal rigidities and invoicing, from capital, intermediates and supply chains, from trade imbalances and asset trade, and from risk and policy uncertainty. We do not attempt to fit the 2025 data. The point of the paper is conceptual: one cannot appeal to the two-country prediction to conclude that an observed multilateral depreciation must reflect financial or non-trade forces. Conversely, the framework says when such an attribution *is* warranted.

1.1 Related literature

A rapidly growing literature studies the exchange-rate effects of the 2025 tariffs. One strand emphasizes financial channels: tariff uncertainty and risk premia (Kalemli-Özcan et al., 2026), the erosion of Treasury convenience yields (Acharya and Laarits, 2026; Jiang et al., 2025), and valuation effects on dollar liabilities (Itskhoki and Mukhin, 2025). A second strand documents that the dollar's response depends on retaliation (Ostry et al., 2025), interpreted through dominant-currency pricing. A third quantifies the welfare and third-country effects of tariffs in rich quantitative models (Ignatenko et al., 2025; Darracq Pariès et al., 2026; Fajgelbaum and Khandelwal, 2026). What is missing is a characterization of what the real trade channel *itself* implies for bilateral and effective exchange rates once trade is multilateral. We provide that characterization. It shows that the configuration-dependence documented by Ostry et al. (2025) arises in a frictionless real model, and it identifies when observed movements, such as the late-2025 dollar weakness under nearly uniform tariffs, cannot be attributed to trade.

Tariffs and exchange rates. The classical two-country analysis of tariffs under flexible exchange rates (Lerner, 1936; Metzler, 1949; Mundell, 1961; Dornbusch, 1974; Eichengreen, 1981; Krugman, 1982) delivers the appreciation result; Dornbusch (1974) is particularly close

to us in giving non-traded goods a role in the incidence of a tariff, which enters our domestic diversion margin through the rebate. Costinot and Werning (2019) give the modern statement of Lerner symmetry; since our results concern the real allocation, they are unaffected by Lerner-type reformulations of the policy. A large open-macroeconomy literature studies tariffs in the short run with nominal rigidities, where monetary policy, invoicing and the fiscal use of revenue shape the response (Lindé and Pescatori, 2019; Barattieri et al., 2021; Erceg et al., 2023; Jeanne, 2021; Bergin and Corsetti, 2023; Auray et al., 2025; Bianchi and Coulibaly, 2025; Werning et al., 2025). We complement that work with the long-run real benchmark in a multilateral setting, the position toward which those short-run dynamics converge once prices have adjusted.

The 2025 dollar. The financial and uncertainty explanations of the 2025 dollar (Kalemli-Özcan et al., 2026; Acharya and Laarits, 2026; Itskhoki and Mukhin, 2025; Cardani et al., 2025; Jiang et al., 2025) are complements to our analysis, not competitors; Kalemli-Özcan et al. (2026), for instance, work in a two-country model, where tariff levels appreciate the currency and tariff volatility depreciates it. Itskhoki and Mukhin (2025) are the closest theory paper on the dollar: in their static benchmark a tariff appreciates the currency, and with dollar-denominated US liabilities that appreciation carries a valuation cost that sharply lowers the optimal macro tariff; a tariff war can then cost the United States its exorbitant privilege. Their mechanism is valuation and convenience yield; ours is real trade diversion. The paper closest to ours empirically is Ostry et al. (2025). Building a database of tariff announcements and retaliations for 2018–20 and 2025, they show that the dollar appreciates on unilateral US tariffs and depreciates when partners retaliate, while in US–China tit-for-tat exchanges it is roughly flat bilaterally. They interpret the pattern through dominant-currency pricing (Gopinath et al., 2020). Our bilateral-retaliation result predicts exactly the same pattern, with $d\nu_B = 0$ against the retaliating partner and depreciation against bystanders whenever $\rho > \rho_D$, without any nominal friction. The two explanations can coexist. Ours shows that the configuration-dependence they document is what the real trade channel requires in the long run, so it need not be read as evidence of a nominal or financial friction, and it identifies the substitution conditions under which that reading is warranted.

Third-country effects. Because the exchange rate *is* the terms of trade in our model class, the preferential-trade-agreement literature on third-country terms-of-trade effects applies directly (Viner, 1950; Mundell, 1964; Kemp and Wan, 1976; Krugman, 1991; Bond and Syropoulos, 1996; Bagwell and Staiger, 1999, 2002; Yi, 2000; Chang and Winters, 2002;

Ornelas, 2005). Our contribution to it is narrow: the competing-exporter result, in which the excluded exporter's terms of trade always improve, becomes conditional ($\rho > N\rho_D$) once the importer also produces a competing good. Quantitative and empirical work on the third-country effects of trade wars (Darracq Pariès et al., 2026; Ignatenko et al., 2025; Fajgelbaum et al., 2024; Freund et al., 2024; Alfaro and Chor, 2023; Fajgelbaum and Khandelwal, 2026) quantifies these effects in rich environments; Darracq Pariès et al. (2026) conclude that tariff incidence is intrinsically multilateral and cannot be inferred from two-country analysis. These papers quantify; we characterize signs.

Elasticities. Our thresholds turn on the gap between the home–import (macro) and cross-origin (micro) elasticities. The evidence that cross-foreign elasticities exceed home–foreign ones (Broda and Weinstein, 2006; Feenstra et al., 2018; Li et al., 2022), and that long-run trade elasticities are larger than short-run ones (Boehm et al., 2023), is exactly the distinction our results require.

Multi-market stability. The MML condition descends from the Hicks–Metzler–Mosak conditions for the stability of multiple markets and from the gross-substitutes stability results (Hicks, 1939; Mosak, 1944; Metzler, 1945; Arrow and Hurwicz, 1958; Arrow et al., 1959). We apply that lineage to the tariff–exchange-rate mapping rather than inventing the condition; the IMF's multilateral exchange rate model (Artus and Rhomberg, 1973) is an early institutional antecedent of multilateral exchange-rate modelling of this kind.

Roadmap. Section 2 develops the general framework. Section 3 analyzes the symmetric case and its three-country representation. Section 4 turns to asymmetry and the 2025 episode, and Section 5 concludes. Section A collects the linearization, Section B the proofs, and Section C departures from the symmetric free-trade benchmark. Online Section D contains further robustness results.

2 Tariffs and Exchange-Rate Adjustment: A General Framework

2.1 The bilateral case

Consider two countries, A and B. Let e_{AB} denote units of A's currency per unit of B's, so that a rise is a depreciation of A, and write $\nu \equiv \log e_{AB}$. Country A levies an ad valorem

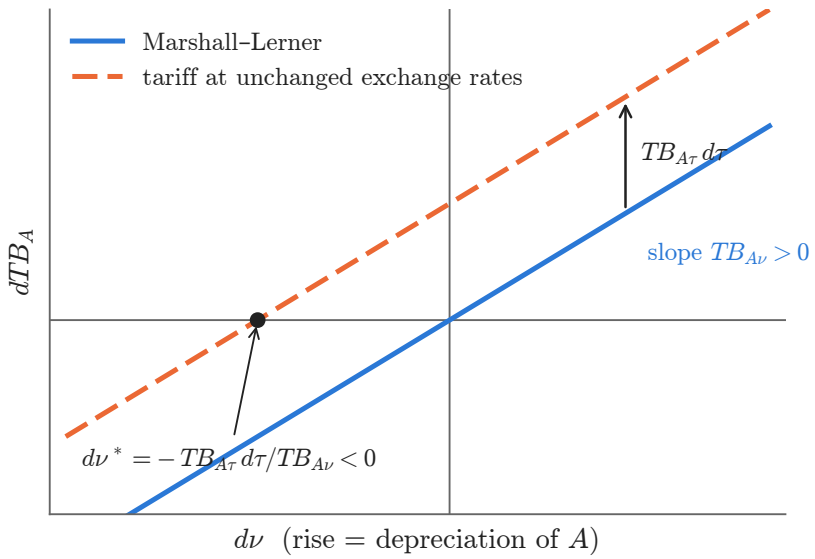


Figure 1: Bilateral tariff and exchange-rate adjustment. The solid line is A's trade balance as a function of the exchange rate at the initial tariff, with slope $\mathrm{TB}_{A\nu} > 0$ under Marshall–Lerner. The tariff shifts it up by $\mathrm{TB}_{A\tau}\, d\tau$ at unchanged exchange rates (dashed line), and balance is restored at $d\nu^* = -\mathrm{TB}_{A\tau}\, d\tau / \mathrm{TB}_{A\nu} < 0$, an appreciation of A. Schematic.

tariff τ on imports from B. Equilibrium with balanced trade is $\mathrm{TB}_A(\nu, \tau) = 0$. Linearizing,

$$d\,\mathrm{TB}_A = \mathrm{TB}_{A\nu}\, d\nu + \mathrm{TB}_{A\tau}\, d\tau = 0 \quad \Longrightarrow \quad d\nu^* = -\frac{\mathrm{TB}_{A\tau}}{\mathrm{TB}_{A\nu}}\, d\tau, \tag{2.1}$$

where $\mathrm{TB}_{A\nu} \equiv \partial \mathrm{TB}_A / \partial \nu$ and $\mathrm{TB}_{A\tau} \equiv \partial \mathrm{TB}_A / \partial \tau$ at fixed ν. A tariff raises the domestic price of imports and, at unchanged exchange rates, improves A's trade balance ($\mathrm{TB}_{A\tau} > 0$). The response is then an appreciation if and only if $\mathrm{TB}_{A\nu} > 0$, that is, if and only if a depreciation improves the trade balance. At a balanced baseline $\mathrm{TB}_{A\nu} = X(\varepsilon_X + \varepsilon_M - 1)$, where X is the value of exports and $\varepsilon_X, \varepsilon_M$ are the export and import demand elasticities, so this is the Marshall–Lerner condition $\varepsilon_X + \varepsilon_M > 1$. Figure 1 illustrates.

Remark 2.1 (Robustness of the two-country result). The sign $d\nu^*/d\tau < 0$ requires only that (i) import expenditure falls with its consumer price and the tariff does not reduce export revenue by more at fixed ν, and (ii) Marshall–Lerner holds. These conditions are satisfied under CES, nested-CES and quasi-linear preferences, under diminishing returns or sector-specific factors with upward-sloping export supply, and under any standard treatment of tariff revenue (Online Section D.1). The correct reading is not that appreciation holds in every model, but that **the two-country result holds wherever Marshall–Lerner holds**. The rest of the paper shows that with three or more countries this is no longer enough.

2.2 The disturbance–adjustment representation

Now consider N countries. Take A's currency as numeraire. The $N-1$ bilateral rates e_{Aj}, $j \neq A$, span all exchange-rate changes, and Walras' law leaves $N-1$ independent trade-balance conditions:

$$\mathrm{TB}_i\Big(\{e_{Aj}\}_{j\neq A}, \{\tau_{jk}\}_{j\neq k}\Big) = 0, \qquad i \neq A.$$

Stack $\nu = (\nu_j)_{j\neq A}$ with $\nu_j = \log e_{Aj}$, and the bilateral tariff changes as $d\tau = (d\tau_{jk})$. Linearizing,

$$J\,d\nu + G\,d\tau = 0, \qquad J_{il} \equiv \frac{\partial \mathrm{TB}_i}{\partial \nu_l}, \qquad G_{i,(jk)} \equiv \left.\frac{\partial \mathrm{TB}_i}{\partial \tau_{jk}}\right|_{\nu}. \tag{2.2}$$

Write a tariff change as a direction a ($\|a\| = 1$) times a scale ds, and let $F \equiv Ga$. Then

$$\boxed{\frac{d\nu}{ds} = (-J)^{-1}F.} \tag{2.3}$$

This separates two conceptually distinct objects. F is the **tariff incidence**: the vector of trade-balance disturbances at unchanged exchange rates, including expenditure switching, tariff revenue and income effects. $(-J)^{-1}$ is the **exchange-rate adjustment**: the movements needed to offset any given vector of disturbances. In the bilateral case both are scalars, and Eq. (2.3) reduces to Eq. (2.1). With $N \geq 3$, F can have entries of different signs, because a tariff on one partner may *improve* another partner's trade balance.

2.3 The multilateral Marshall–Lerner condition

Definition 2.2 (Multilateral Marshall–Lerner condition, MML)**.** The adjustment mechanism satisfies MML if $J_{il} \geq 0$ for $l \neq i$ and $(-J)^{-1}$ exists with $(-J)^{-1} \geq 0$ elementwise. Equivalently, $-J$ is a nonsingular M-matrix.

Under MML, writing $\vartheta_{ij} \equiv [(-J)^{-1}]_{ij} \geq 0$,

$$\frac{d\nu_i}{ds} = \sum_j \vartheta_{ij} F_j. \tag{2.4}$$

A positive disturbance in any country contributes weakly positively to every bilateral response. So **a uniformly signed F cannot generate responses of the opposite sign**, which is the multilateral version of the two-country logic. When F has mixed signs, the response of any bilateral rate depends on the whole vector of disturbances, not on its own entry.

A more primitive sufficient condition uses the full $N \times N$ Jacobian $\tilde{J}_{il} = \partial \mathrm{TB}_i / \partial \nu_l$ in terms of currency values, where now ν_l is the log value of currency l in a common unit of account. Homogeneity of degree one gives $\sum_l \tilde{J}_{il} = \mathrm{TB}_i = 0$ at a balanced baseline, and Walras' law gives $\sum_i \tilde{J}_{il} = 0$.

Assumption 2.1 (Gross substitutability of trade balances)**.** $\tilde{J}_{il} \geq 0$ for all $l \neq i$: an appreciation of currency l weakly improves every other country's trade balance.

Lemma 2.3. *If each trade balance is homogeneous of degree one in currency values, Assumption 2.1 holds, J is irreducible, and $\tilde{J}_{iA} > 0$ for some i, then MML holds.*

This is the trade-balance analogue of the gross-substitutes stability results of Hicks (1939), Mosak (1944), Metzler (1945) and Arrow et al. (1959). We use it as the multilateral counterpart of the Marshall–Lerner condition. The point to stress is that **MML does not pin down the sign of any individual exchange-rate response.** Every bilateral Marshall–Lerner condition can hold and the multilateral response can still go either way. What decides it is the structure of F.

3 The Symmetric Case

3.1 Average protection and relative treatment

Restrict attention to tariff changes imposed by A, $t = (t_j)_{j \neq A}$ with $t_j \equiv d\tau_{Aj}$, and let G now denote the $(N-1) \times (N-1)$ restriction of the tariff-incidence matrix. Suppose A's partners are **symmetric**: $PJP' = J$ and $PGP' = G$ for every permutation P of the partners, as at free trade or under a common external tariff. Decompose

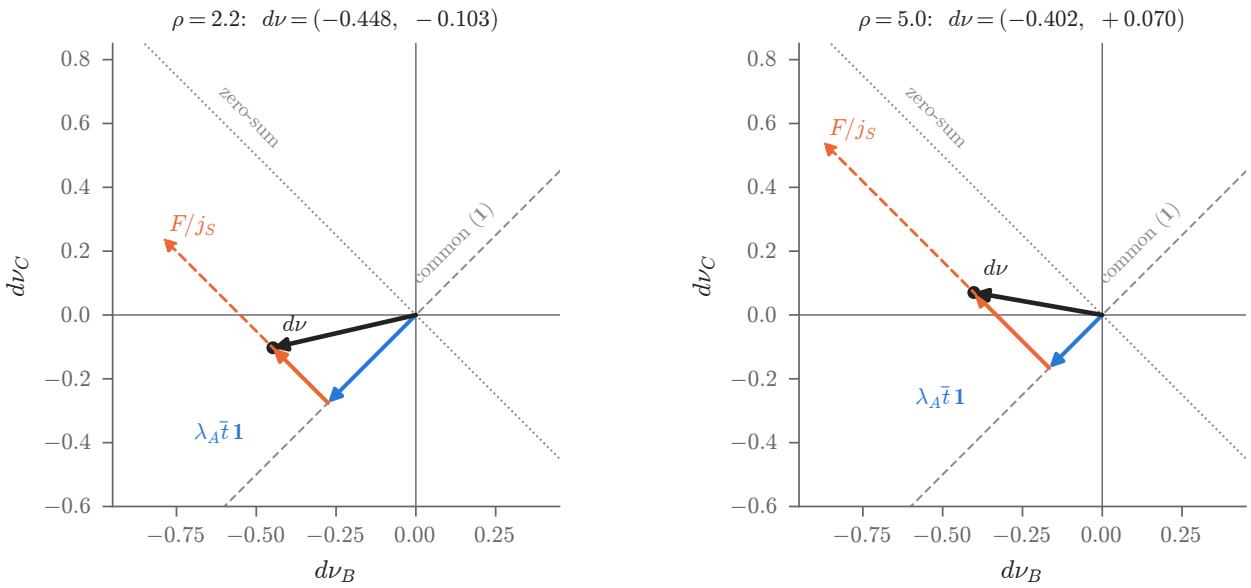
$$t = \bar{t}\,\mathbf{1} + \tilde{t}, \qquad \bar{t} = \tfrac{1}{N-1}\mathbf{1}'t, \qquad \mathbf{1}'\tilde{t} = 0. \tag{3.1}$$

We call $\bar{t}$ **average protection** and $\tilde{t}$ **relative treatment**. Any permutation-invariant matrix has the form $aI + b\mathbf{1}\mathbf{1}'$, so $\mathbf{1}$ and every zero-sum vector are eigenvectors of both G and $-J$:

$$G\mathbf{1} = g_S\mathbf{1}, \quad Gv = g_D v; \qquad (-J)\mathbf{1} = j_S\mathbf{1}, \quad (-J)v = j_D v \qquad (\mathbf{1}'v = 0).$$

Proposition 3.1 (Average protection and relative treatment)**.** *Let partners be symmetric, $N \geq 3$, and $j_S, j_D \neq 0$. For any tariff vector t,*

$$d\nu = \lambda_A\,\bar{t}\,\mathbf{1} + \lambda_R\,\tilde{t}, \qquad \lambda_A = g_S/j_S, \quad \lambda_R = g_D/j_D. \tag{3.2}$$



(a) $\rho = 2.2$: $\rho_D < \rho < 3\rho_D$ **(b)** $\rho = 5$: $\rho > 3\rho_D$

Figure 2: The geometry of average protection and relative treatment ($N = 3$). A unilateral tariff $t = (1, 0) = (\frac{1}{2}, \frac{1}{2}) + (\frac{1}{2}, -\frac{1}{2})$, with $\alpha_D = 0.7$, $\alpha_T = 0.6$, $\eta = 1.5$ ($\rho_D = 1.17$, $3\rho_D = 3.51$). Blue: common component $\lambda_A \bar{t}\,\mathbf{1}$; solid orange: relative component $\lambda_R \tilde{t}$; black: equilibrium $d\nu$. The dashed orange arrow scales the relative component by N to show F/j_S. In (a), $\rho_D < \rho < 3\rho_D$: C's trade balance improves at unchanged exchange rates ($F_C > 0$), yet A still appreciates against C. In (b), $\rho > 3\rho_D$: A depreciates against C. In both, the NEER appreciates. Computed from Theorem 3.2.

Hence, with $d\nu_A^E \equiv \frac{1}{N-1}\sum_{j \neq A} d\nu_j$ *(the NEER at symmetric weights),*

$$d\nu_A^E = \lambda_A \bar{t}, \qquad d\nu_j - d\nu_A^E = \lambda_R \tilde{t}_j, \qquad d\log e_{jk} = \lambda_R (t_k - t_j). \tag{3.3}$$

Moreover, MML is equivalent to $j_S > 0$ *and* $j_D \geq j_S$.

Average protection alone moves the NEER; relative treatment alone moves bilateral rates around it and the cross rates among partners. The geometry is simple. The response vector lives in the plane spanned by the common direction $\mathbf{1}$ and the zero-sum directions, and each component is scaled by its own coefficient (Fig. 2).

3.2 A nested-CES economy

Each country $i \in \{1, \ldots, N\}$ produces a country-specific tradable and a non-tradable with labor only, $Y_{Ti} = A_{Ti} L_{Ti}$ and $Y_{Ni} = A_{Ni} L_{Ni}$. Normalizing each tradable's producer price to one in its own currency, $w_i = A_{Ti}$ and $P_{Ni} = A_{Ti}/A_{Ni}$. Relative international producer prices therefore move only through exchange rates, and **the bilateral exchange rate equals the**

bilateral terms of trade and the relative wage in a common currency.

Preferences are

$$C_i = C_{Ti}^{\alpha_T} C_{Ni}^{1-\alpha_T}, \quad C_{Ti} = \left[\alpha_D^{1/\eta} C_{Tii}^{\frac{\eta-1}{\eta}} + (1-\alpha_D)^{1/\eta} M_i^{\frac{\eta-1}{\eta}}\right]^{\frac{\eta}{\eta-1}}, \quad M_i = \left[\sum_{j\neq i} \beta_{ij}^{1/\rho} C_{Tji}^{\frac{\rho-1}{\rho}}\right]^{\frac{\rho}{\rho-1}}. \tag{3.4}$$

Here η governs substitution between domestic tradables and the import composite, and ρ substitution across foreign origins; the latter is inactive when $N = 2$. Country i levies τ_{ij} on imports from j, so the consumer price is $p_{ij} = (1+\tau_{ij})e_{ij}$, and the import and tradable price indices are $P_{Mi} = [\sum_{j\neq i} \beta_{ij} p_{ij}^{1-\rho}]^{1/(1-\rho)}$ and $P_{Ti} = [\alpha_D + (1-\alpha_D) P_{Mi}^{1-\eta}]^{1/(1-\eta)}$. Tariff revenue is rebated lump-sum, so income is

$$I_i = \frac{w_i L_i}{1 - \sum_{j\neq i} \frac{\tau_{ij}}{1+\tau_{ij}} s_{ij}}, \tag{3.5}$$

with s_{ij} the realized tariff-inclusive expenditure share of income on j's good. Trade balances are measured at border prices,

$$\mathrm{TB}_i = \sum_{k\neq i} f_{ki} - \sum_{j\neq i} f_{ij}, \qquad f_{ij} = \frac{s_{ij}\tilde{I}_i}{1+\tau_{ij}}, \tag{3.6}$$

with $\tilde{I}_i = e_{Ai} I_i$ income in A's currency. The NEER uses baseline bilateral trade weights, $E_i = \prod_{j\neq i} e_{ij}^{w_{ij}}$ with $w_{ij} = (f_{ij} + f_{ji}) / \sum_k (f_{ik} + f_{ki})$; throughout, "effective exchange rate" refers to this trade-weighted NEER. The real exchange rate is $q_{ij} = e_{ij} P_j / P_i$, with P_i the consumer price index.

The **symmetric benchmark** is free trade with $\beta_{ij} = 1/(N-1)$, common parameters, and equal endowments and productivities. It retains home bias α_D. Write $m \equiv \alpha_T(1-\alpha_D)$ for the import share of income.

The two-country case. With one foreign supplier, linearizing gives $d\mathrm{TB}_A = m M_2\, d\nu_B + m\rho_D\, d\tau$, so

$$\left.\frac{d\log e_{AB}}{d\tau}\right|_{\tau=0} = -\frac{\rho_D}{M_2}, \qquad \rho_D \equiv \underbrace{\alpha_D \eta}_{\text{domestic substitution}} + \underbrace{(1-\alpha_D)(1-\alpha_T)}_{\text{rebate / non-tradables}}, \quad M_2 \equiv 1 + 2\alpha_D(\eta-1) = \varepsilon_X + \varepsilon_M - 1. \tag{3.7}$$

ρ_D is the **domestic diversion margin**. It measures how strongly a tariff pulls expenditure toward the domestic economy, through substitution toward the domestic tradable and through the rebate's income effect on non-tradables. An exchange-rate movement generates

no revenue, so it has no rebate term. This is why $\rho_D \neq M_2$.

3.3 The main theorem

Theorem 3.2 (Exchange-rate responses in the symmetric nested-CES model)**.** *At the symmetric benchmark define*

$$M_N \equiv \frac{1}{N-1}\Big[1 + \underbrace{N\alpha_D(\eta-1)}_{\textit{domestic–import adjustment}} + \underbrace{(N-2)\rho}_{\textit{foreign-origin adjustment}}\Big]. \tag{3.8}$$

MML holds if and only if $M_N > 0$, and $\eta \geq 1$ suffices. When it holds,

$$\lambda_A = -\frac{\rho_D}{M_N}, \qquad \lambda_R = -\frac{\rho}{NM_N}, \qquad d\nu_j = \lambda_A \bar{t} + \lambda_R \tilde{t}_j. \tag{3.9}$$

M_N measures how sensitive A's trade balance with the rest of the world is to its effective exchange rate: $d\mathrm{TB}_A = mM_N\, d\nu_A^E$. When $N > 2$ it includes ρ, because foreign buyers can substitute between A's goods and other foreign goods. For $N = 2$ it reduces to M_2. Average protection works as in the two-country case. Relative treatment is zero-sum, so to first order it leaves revenue, income and the tradable/non-tradable split unchanged, and its disturbance is governed by ρ alone.

The factor $1/N$ in λ_R is central. A common exchange-rate move shifts only partners' trade with A. A relative move also shifts partners' trade with *each other*. At the symmetric benchmark A accounts for a share $1/N$ of the trade on which relative movements act, so $j_D = Nj_S$. **Relative exchange-rate adjustment is N times as effective as common adjustment**, and a given relative disturbance requires only $1/N$ as much exchange-rate movement.

3.4 A unilateral tariff

Let A raise its tariff on B alone, $t = d\tau\, e_B$. Then $\bar{t} = d\tau/(N-1)$, $\tilde{t}_B = \frac{N-2}{N-1}d\tau$ and $\tilde{t}_C = -\frac{d\tau}{N-1}$ for every bystander C.

Proposition 3.3 (Unilateral tariff)**.** *At the symmetric benchmark under MML,*

$$\frac{d\nu_A^E}{d\tau} = -\frac{\rho_D}{(N-1)M_N} < 0, \qquad \frac{d\nu_B}{d\tau} = -\frac{N\rho_D + (N-2)\rho}{N(N-1)M_N} < 0, \qquad \frac{d\nu_C}{d\tau} = \frac{\rho - N\rho_D}{N(N-1)M_N}. \tag{3.10}$$

(i) A's NEER appreciates for every N and ρ.

(ii) A appreciates against the target.

(iii) *A* ***depreciates against each untariffed bystander if and only if*** $\rho > N\rho_D$.

(iv) *The same threshold applies when A tariffs any* $k \leq N-2$ *partners. More generally, A depreciates against a partner j facing a below-average tariff increase if and only if* $t_j/\bar{t} < 1 - N\rho_D/\rho$, *which requires* $\rho > N\rho_D$.

Against the target, average protection and relative treatment reinforce each other. Against a bystander they pull apart. Average protection shifts expenditure toward the domestic economy (appreciation), while the bystander's favorable relative treatment redirects expenditure toward it (depreciation).

Corollary 3.4 (Why a single elasticity cannot deliver reversal)**.** *With home bias* $\alpha_D \geq 1/N$,

$$N\rho_D - \eta = \eta(N\alpha_D - 1) + N(1-\alpha_D)(1-\alpha_T) \geq 0, \tag{3.11}$$

so $\rho > \eta$ *is* ***necessary*** *for a bystander depreciation. In any specification with a single elasticity across all tradable varieties* ($\rho = \eta = \sigma$)*, A appreciates against every bystander at every* σ. *At* $\alpha_D = 1/N$ *the response is* $-(1-\alpha_T)/(N\sigma)$, *which approaches zero from below but never crosses it.*

This explains why the natural first extension of the two-country model, a third country under a single CES elasticity, has not surfaced the reversal. The reversal requires that cross-origin substitution exceed home–import substitution, an ordering supported by the macro–micro elasticity evidence (Feenstra et al., 2018; Li et al., 2022). At US-like parameters ($\alpha_D = 0.8$, $\alpha_T = 0.4$, $\eta = 1.5$, so $\rho_D = 1.32$), $N\rho_D = 3.96$ for $N = 3$.

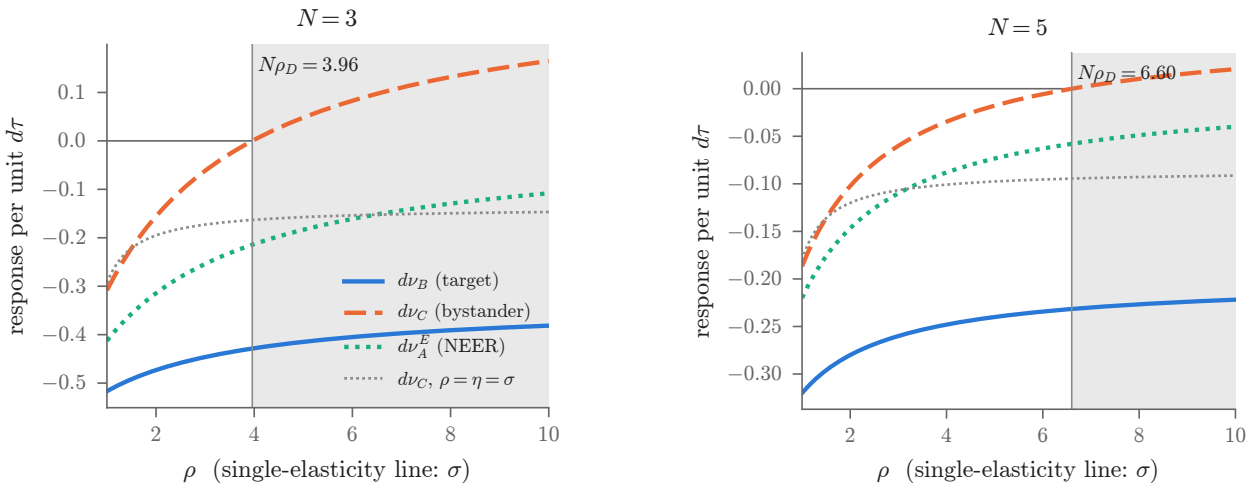
Disturbance versus equilibrium. At unchanged exchange rates the bystander's trade balance improves if and only if $\rho > \rho_D$:

$$F_C = \frac{m(\rho - \rho_D)}{(N-1)^2}. \tag{3.12}$$

The equilibrium bystander depreciation requires the stronger condition $\rho > N\rho_D$. The gap is the $1/N$ effectiveness of relative adjustment. For $\rho_D < \rho < N\rho_D$, C gains at the old exchange rates yet ends up with a weaker currency against A (Fig. 2a). Figure 3 plots the three responses against ρ.

3.5 Retaliation and trade wars

Since the analysis is local, simultaneous tariff changes by several countries add up.



(a) $N = 3$ **(b)** $N = 5$

Figure 3: Bilateral and effective responses to a unilateral tariff, per unit $d\tau$, US-like parameters ($\alpha_D = 0.8$, $\alpha_T = 0.4$, $\eta = 1.5$). $d\nu_B$ and $d\nu_A^E$ are negative for every ρ; $d\nu_C$ crosses zero at $N\rho_D$ (3.96 for $N = 3$, 6.60 for $N = 5$), and the shaded region is $\rho > N\rho_D$. The grey dotted line is the single-elasticity case $\rho = \eta = \sigma$, plotting $d\nu_C$ against σ; it never crosses zero (Corollary 3.4). Closed forms of Proposition 3.3, verified against the linear solver.

Proposition 3.5 (Retaliation and trade wars)**.** *At the symmetric benchmark with $N \geq 3$ under MML:*

(i) ***Bilateral retaliation*** *(A tariffs B and B tariffs A by $d\tau$): $d\nu_B = 0$, and*

$$\frac{d\nu_C}{d\tau} = \frac{\rho - \rho_D}{(N-1)M_N}, \qquad \frac{d\nu_A^E}{d\tau} = \frac{N-2}{N-1} \cdot \frac{\rho - \rho_D}{(N-1)M_N}. \tag{3.13}$$

Both belligerents depreciate against every bystander, and in effective terms, ***if and only if*** *$\rho > \rho_D$. Cross rates among bystanders are unchanged.*

(ii) ***Multilateral trade war*** *(A tariffs every partner, and every partner retaliates against A): $d\nu_j = d\nu_A^E = \frac{(N-2)(\rho-\rho_D)}{(N-1)M_N}\, d\tau$ for all j, so A depreciates against everyone if and only if $\rho > \rho_D$.*

B's retaliation adds a second relative-treatment effect in the bystander's favor, which offsets the $1/N$ dilution. The threshold drops from $N\rho_D$ to ρ_D, the point at which the bystander's trade balance improves at unchanged exchange rates. At US-like parameters $\rho_D = 1.32$, below essentially all cross-origin estimates. **Retaliation therefore makes multilateral depreciation of the tariffing country's currency the generic outcome, while leaving the belligerents' bilateral rate unchanged.** This is the pattern documented by Ostry et al. (2025), including the near-stability of the dollar against the renminbi

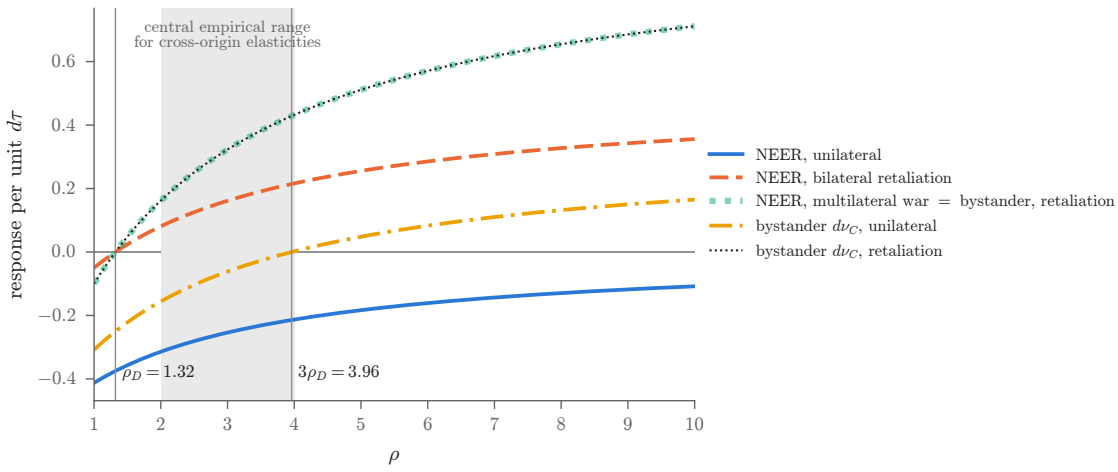


Figure 4: Configuration dependence, $N = 3$, US-like parameters. NEER responses $d\nu_A^E/d\tau$ under a unilateral tariff (always negative), bilateral retaliation and a multilateral war (both crossing zero at $\rho_D = 1.32$), and the bilateral response against a bystander under a unilateral tariff (crossing at $3\rho_D = 3.96$) and under retaliation (crossing at ρ_D). At $N = 3$ the multilateral-war response coincides with the bystander response under bilateral retaliation, so those two lines overlap. Shaded: the central empirical range for cross-origin elasticities, $\rho \in [2, 4]$.

in tit-for-tat episodes. Here it arises with flexible prices, balanced trade and no financial channel.

The two experiments differ in where adjustment falls. Under bilateral retaliation the belligerents' rate is fixed and adjustment runs against bystanders. In a multilateral war there are no bystanders, and all of A's rates move together (Fig. 4).

3.6 The three-country representation

The N-country results can be read through a three-country economy, which we use as the workhorse for intuition. When the $N - 2$ bystanders are symmetric and the tariff shock preserves that symmetry, no relative-price movement within the bystander bloc is activated. The bloc then has a single active direction, and the tariff experiment reduces exactly to three regions: A, B and a composite bystander C. We now show that this composite is itself a nested-CES country, and we identify its parameters.

Proposition 3.6 (Exact three-country reduction)**.** *At the symmetric benchmark, consider any tariff change among A, B and the bystanders that is common across bystanders: tariffs by A or B on all bystanders, by all bystanders on A or B, or between A and B. To first order, the responses of ν_B, of every bystander rate ν_C and of A's NEER coincide with those of a three-country nested-CES economy $\{A, B, C\}$ in which:*

- *C has income $y_C = N - 2$;*

- *A and B place import weight $\frac{N-2}{N-1}$ on C and $\frac{1}{N-1}$ on each other;*
- *C places import weight $\frac{1}{2}$ on each of A and B, with cross-origin elasticity ρ;*
- *C's home share of tradables is $h_C = \alpha_D + (1-\alpha_D)\frac{N-3}{N-1}$, equivalently $(N-1)h_C = N-3+2\alpha_D$;*
- *C's home–import elasticity is*

$$\boxed{\eta_C = \frac{(N-3)\rho + 2\alpha_D\,\eta}{N-3+2\alpha_D}.} \tag{3.14}$$

Proof sketch. All bystander goods have the same price change, so CES aggregation of A's and B's demand over bystander goods is exact. Trade within the bloc cancels from the bloc's trade balance; what remains is the bloc's demand for A's and B's goods. Measure prices in bystander currency, so bloc goods have $dp = 0$, and write $S = dp_A + dp_B$ and $D = dp_A - dp_B$. A bystander's import price index moves by $S/(N-1)$, since A and B are 2 of its $N-1$ import origins. Hence

$$d\log s_{cA} = (1-\rho)\tfrac{D}{2} + \left[(1-\rho)\tfrac{N-3}{2(N-1)} + (1-\eta)\tfrac{\alpha_D}{N-1}\right]S.$$

In the composite the import index moves by $S/2$, so $d\log s_{CA} = (1-\rho)\frac{D}{2} + (1-\eta_C)h_C\frac{S}{2}$. The D terms coincide. Matching the S terms gives $(1-\eta_C)(N-1)h_C = (1-\rho)(N-3)+2\alpha_D(1-\eta)$, which is Eq. (3.14). The full proof is in Section B. □

The reduction is first-order. At finite tariffs a CES with constant η_C does not replicate the bloc.

Interpretation. The composite bystander differs from an ordinary large country in one respect. Most of each bystander's trade is with *other bystanders*, whose goods sit in the same ρ-nest as A's and B's. Once collapsed, this internal trade becomes the composite's "home" trade, and it is substitutable at a blend of ρ and η. When $\rho > \eta$, η_C rises toward ρ as N grows. The bystander bloc becomes inward-looking ($h_C \to 1$) and highly substitutable.

The reduced system. In the composite economy the 2×2 system for (ν_B, ν_C) is, with $c = mM_N/(N-1)$,

$$-J = c\begin{pmatrix} N-1 & -(N-2) \\ -(N-2) & 2(N-2) \end{pmatrix}, \qquad \det(-J) = c^2 N(N-2), \tag{3.15}$$

Table 1: N versus bystander size

		Composite bystander threshold		
N	$N\rho_D$ (true)	with η_C Eq. (3.14)	with ordinary η	η_C at $\rho = 4$
3	3.96	3.96	3.96	1.50
4	5.28	5.28	3.82	2.46
5	6.60	6.60	3.76	2.89
8	10.56	10.56	3.71	3.39

Notes: US-like parameters ($\alpha_D = 0.8$, $\alpha_T = 0.4$, $\eta = 1.5$). Thresholds are the values of ρ at which the bystander response to a unilateral tariff changes sign in the composite economy of Proposition 3.6, computed with the linear system of Section A. The "ordinary η" column keeps the composite's flows but replaces η_C by η.

and

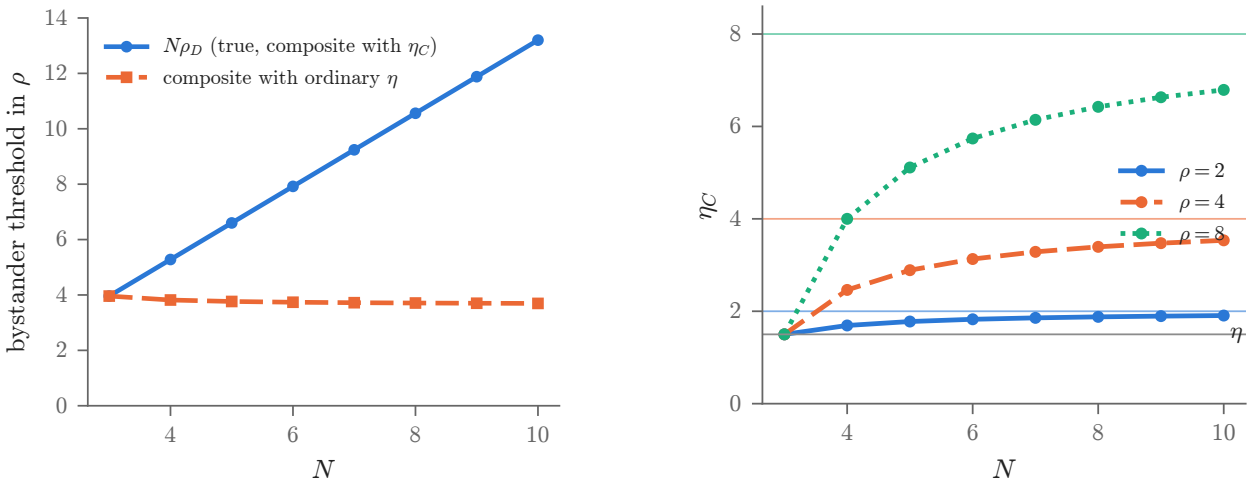
$$d\nu_C = \frac{F_B + (N-1)F_c}{cN}, \qquad d\nu_B = \frac{2F_B + (N-2)F_c}{cN}, \tag{3.16}$$

where F_B is the target's disturbance and F_c each bystander's ($F_C = (N-2)F_c$). Under a unilateral tariff, $F_B = -m[\rho_D + (N-2)\rho]/(N-1)^2 < 0$ and $F_c = m(\rho - \rho_D)/(N-1)^2$. Equation (3.16) recovers Eq. (3.10) and gives the bystander response a two-force reading:

- **diversion**, $(N-1)F_c$: expenditure diverted toward bystanders improves their position and strengthens their currencies;
- **redirection**, F_B: the target loses access to A's market and its currency must fall. This makes the target's goods cheaper for bystanders and drags bystander currencies down with it.

A depreciates against the bystander if and only if diversion outweighs redirection, $(N-1)(\rho - \rho_D) > \rho_D + (N-2)\rho$, which is $\rho > N\rho_D$. These are the "trade diversion" and "trade redirection" forces of earlier versions of this paper, now with exact weights.

Why N raises the threshold, and why size alone does not. One might expect that adding countries is equivalent to enlarging the bystander. It is not. Hold the composite's flows fixed at their N-country values but give it an ordinary home–import elasticity η instead of η_C. The bystander threshold then *falls* slightly with N, consistent with the general result (Section C.3) that a larger bystander lowers the threshold. The true threshold $N\rho_D$ rises steeply (Table 1 and Fig. 5). What drives the N-dependence is the bloc's internal, highly substitutable trade, which makes relative exchange-rate adjustment N times as effective as common adjustment and dilutes A's footprint to $1/N$.



(a) Bystander threshold **(b)** Effective elasticity η_C

Figure 5: The composite bystander, US-like parameters. (a) The true threshold $N\rho_D$ for $N = 3, \ldots, 10$, which the composite with η_C reproduces exactly, and the threshold of the "ordinary-η" composite, which is nearly flat and slightly declining. (b) η_C from Eq. (3.14) against N for $\rho \in \{2, 4, 8\}$, converging toward ρ (thin horizontal lines); the grey line is $\eta = 1.5$.

3.7 Real exchange rates

Real rates add a price-index wedge: the tariff raises A's consumer prices directly. With $\bar{t}$ the average tariff and partners imposing no tariffs,

$$d\log q_{Aj} = \left(1 - \tfrac{mN}{N-1}\right) d\nu_j - m\bar{t}, \qquad dq_A^E = \left[\left(1 - \tfrac{mN}{N-1}\right)\lambda_A - m\right]\bar{t}. \tag{3.17}$$

Proposition 3.7 (Real exchange rates)**.** *Consider a unilateral tariff at the symmetric benchmark.*

(i) If $m < 1/N$ and MML holds at the relevant ρ, the real bilateral threshold exceeds the nominal one:

$$\rho_q^*(N) = \frac{[(N-1) - mN]N\rho_D + mN[N\alpha_D(\eta - 1) + 1]}{(N-1)(1 - mN)} > N\rho_D.$$

(ii) If $m < (N-1)/N$, the REER appreciates.

At US-like parameters and $N = 3$, $\rho_q^* = 4.93$, against $3\rho_D = 3.96$.

3.8 Relation to the competing-exporter model

In our model class the bystander's exchange rate is its terms of trade: $\mathrm{ToT}_{CA} = e_{AC}$. The response in Eq. (3.10) is therefore the third-country terms-of-trade effect of a discriminatory

tariff, the central object of the preferential-trade-agreement literature since Viner (1950) and Mundell (1964). In the canonical competing-exporter model, an importer A that does not produce the good buys it from B and C under quasi-linear preferences. With q_j exporter j's price, E_j its export supply and M_A the importer's demand, a tariff on B then *always* improves C's terms of trade, since

$$\frac{dq_C}{dt_B} = \frac{E'_B}{E'_B + E'_C - M'_A} > 0,$$

and preferential liberalization toward B always hurts the excluded country. In our model the same effects hold **only if** $\rho > N\rho_D$. The difference is that A produces a competing tradable. Part of the demand diverted from B goes to A's own good, along with the rebate's income effect, so the third-country effect survives only when cross-origin substitution is strong relative to domestic diversion. The two environments are not formally nested. The competing-exporter model can be read as switching off ρ_D in order to isolate substitution across foreign suppliers. Section C.1 shows that at a positive common external tariff the threshold rises, and that a discrete preferential cut requires a higher ρ than the local threshold suggests.

4 Asymmetry

The symmetric case delivers clean geometry: the NEER and the bilateral rates separate exactly, and thresholds are linear in ρ. Asymmetry breaks the eigen-structure. This section does not aim at a general theory of asymmetric trade networks. Its purpose is to show which intuitive conclusions of the symmetric case are structural and which are artefacts of symmetry. The most consequential casualty is the result that a unilateral tariff always improves the tariffing country's aggregate terms of trade.

Two weighted averages of bilateral responses appear in this section and should be kept apart. The *effective exchange rate* is, as throughout the paper, the trade-weighted NEER, with weights w_{Aj} that combine A's imports from and exports to j. The *aggregate terms of trade* weight the same bilateral responses by A's import shares b_{Aj}. At the symmetric benchmark the two weightings coincide; away from it they do not, and it is the import-weighted aggregate terms of trade that is welfare-relevant.

4.1 The aggregate terms of trade and welfare

Let b_{Aj} be partner j's share in A's import bundle, and $R_j^B \equiv \partial \nu_j / \partial \tau_{AB}$. Define the import-weighted foreign-price response

$$R_{A|B} \equiv \sum_{j \neq A} b_{Aj} R_j^B. \tag{4.1}$$

Under the producer-price normalization, a rise in e_{Aj} raises the price of j's good relative to A's, so A's aggregate terms of trade improve if and only if $R_{A|B} < 0$.

Lemma 4.1 (First-order welfare)**.** *At a free-trade baseline, for any tariff vector, the first-order change in A's welfare, in units of income, is*

$$\frac{dU_A}{U_A' I_A} = -\, m_A \sum_{j \neq A} b_{Aj}\, d\nu_j, \tag{4.2}$$

where m_A is A's import share of income. A marginal tariff thus raises A's welfare if and only if it improves A's aggregate terms of trade.

Proof sketch. A's wage and non-tradable price are fixed in A's currency. By the envelope theorem, $\lambda\, dU_A = dT_A^{rev} - \sum_j C_{Aj}\, dp_{Aj}$. At free trade $dT_A^{rev} = \sum_j f_{Aj}\, d\tau_{Aj}$ and $dp_{Aj} = p_{Aj}(d\nu_j + d\tau_{Aj})$. The tariff terms cancel, leaving $-\sum_j f_{Aj} d\nu_j$. Foreign tariffs affect A only through $d\nu$. The full proof is in Section B. □

At the symmetric benchmark import weights equal trade weights, so $R_{A|B} = d\nu_A^E / d\tau < 0$ (Proposition 3.3). **A unilateral tariff always improves the tariffing country's aggregate terms of trade and raises its welfare to first order, even when it depreciates against every bystander.** Under bilateral retaliation the sign flips when $\rho > \rho_D$. For $N = 3$,

$$\frac{R_B^B + R_C^B}{2} = -\frac{\rho_D}{2M_3}:$$

the favorable move against the target always dominates the adverse move against the bystander when weights are equal.

4.2 The aggregate terms-of-trade condition

Separate the target from the remaining suppliers, and let $\bar{R}_{-B}^B = \sum_{j \neq A,B} b_{Aj} R_j^B / (1 - b_{AB})$ be the conditional import-weighted bystander response. Then $R_{A|B} = b_{AB} R_B^B + (1 - b_{AB}) \bar{R}_{-B}^B$.

Definition 4.2 (Aggregate terms-of-trade condition)**.** Unilateral tariffs by A have the conventional aggregate terms-of-trade effect if, for every target B,

$$b_{AB} R_B^B + (1 - b_{AB}) \bar{R}_{-B}^B < 0. \tag{4.3}$$

If $R_B^B < 0$ and $\bar{R}_{-B}^B > 0$, the condition fails if and only if

$$\frac{\bar{R}_{-B}^B}{-R_B^B} > \frac{b_{AB}}{1 - b_{AB}}. \tag{4.4}$$

The left side is equilibrium incidence: the adverse bystander move relative to the favorable move against the target. The right side is exposure: A's import dependence on the target relative to other suppliers. A bilateral bystander reversal is therefore **necessary but not sufficient** for an aggregate reversal. The adverse moves must also be large enough and carry enough weight in A's basket.

The aggregate condition always reduces to a three-country form, with one weight $b_{AC} = 1 - b_{AB}$ and one aggregate response. This reduction is exact for *any* configuration of bystanders. What fails under heterogeneous bystanders is the representation of the bloc as an ordinary country with primitive parameters (Section 4.4).

4.3 Asymmetric partners in the three-country model

In the three-country model the local system is $A(\rho)R = F(\rho)$ with $A \equiv -J$. At a given baseline every entry of A and F is affine in ρ, because ρ enters share responses only through $(1 - \rho)$. Since A is 2×2, its adjugate is affine too, and

$$R_{A|B}(\rho) = \frac{b' \operatorname{adj}(A(\rho))F(\rho)}{\det A(\rho)}, \qquad b' \operatorname{adj}(A(\rho))F(\rho) = q_2\rho^2 + q_1\rho + q_0 \equiv Q_{A|B}(\rho). \tag{4.5}$$

Under MML $\det A > 0$, so the aggregate condition is $Q_{A|B}(\rho) < 0$. The same structure governs the bilateral bystander response, whose numerator has leading coefficient

$$\alpha_T^2\, b_{AB} b_{AC} b_{CA} b_{CB} (1 - h_A)(1 - h_C) y_A y_C > 0$$

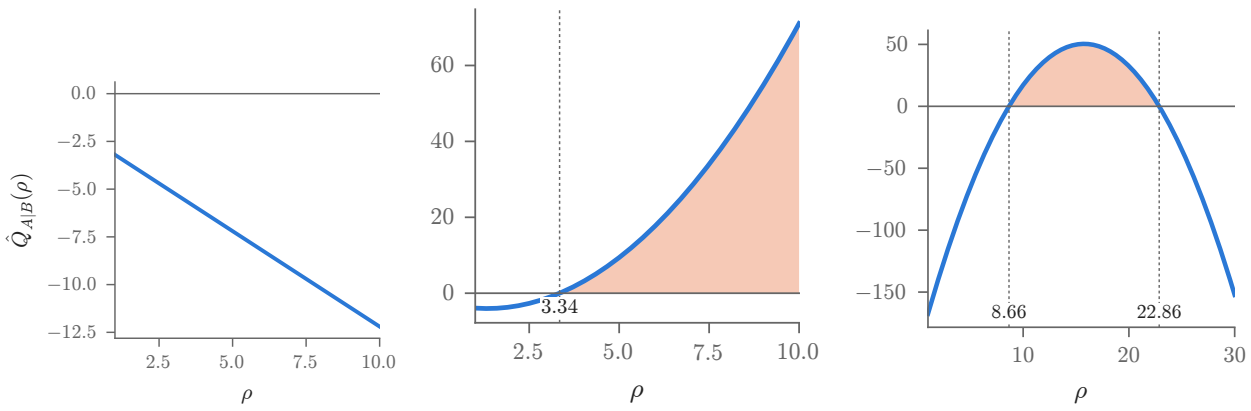
(with a common α_T), where h_i are home shares of tradable spending and y_i incomes.

Proposition 4.3 (Three qualitative cases)**.** *Holding the baseline fixed:*

(a) ***Symmetric:*** *$q_2 = 0$ and $Q < 0$ for all ρ. The condition always holds.*

(b) *$q_2 > 0$,* *$q_0 < 0$****:*** *exactly one positive root ρ^*. The condition holds for $\rho < \rho^*$ and fails for $\rho > \rho^*$.*

(c) *$q_2 < 0$,* *$q_1 > 0$,* *$q_0 < 0$,* ***positive discriminant:*** *two positive roots. The condition fails only for intermediate ρ.*



(a) Symmetric **(b)** Small, low-weight target **(c)** Asymmetric network

Figure 6: Three relationships between ρ and the aggregate terms-of-trade condition. Each panel plots the normalized numerator $\hat{Q}_{A|B}(\rho)$ of Eq. (4.5) at a fixed baseline; the condition fails where $\hat{Q} > 0$ (shaded). (a) Symmetric benchmark: never crosses. (b) Example (b): single crossing at 3.34. (c) Example (c): crossings at 8.66 and 22.86. Signs verified against the linear solver.

Examples. All use $\alpha_D = 0.8$, $\alpha_T = 0.4$, $\eta = 1.5$ unless stated, with the free-trade baseline computed from primitives at $\bar{\rho} = 3\rho_D = 3.96$ and then held fixed; $\hat{Q}$ denotes Q normalized by $|q_2|$ (or by $|q_1|$ when $q_2 = 0$).

(a) Symmetric benchmark: $\hat{Q} = -(\rho + 2.2)$.

(b) Sizes $(S_A, S_B, S_C) = (1, 0.22, 1)$, $\beta_{AB} = 0.2$, $\beta_{AC} = 0.8$, other origin weights symmetric: $\hat{Q} = \rho^2 - 2.6695\rho - 2.2568$, so $\rho^* = 3.344$.

(c) A strongly asymmetric network, specified directly in baseline shares: $\alpha_T = 0.862$, $\eta = 3.007$, home shares $(0.201, 0.854, 0.360)$, origin shares $(b_{AB}, b_{AC}) = (0.525, 0.475)$, $(b_{BA}, b_{BC}) = (0.766, 0.234)$, $(b_{CA}, b_{CB}) = (0.407, 0.593)$, and incomes $(1, 5.126, 0.867)$, which balance trade at these shares. Then $\hat{Q} = -\rho^2 + 31.53\rho - 198.04$, so the condition fails for $8.66 < \rho < 22.86$.

Which asymmetries matter. Fix $S_A = S_C = 1$ and symmetric foreign-origin preferences for B and C. Vary the target's relative size S_B/S_C and its weight β_{AB} in A's imports, recomputing the free-trade equilibrium at $\bar{\rho} = 3\rho_D = 3.96$. In this family $q_0 < 0$ throughout and the two-root case never appears. The sign of q_2 therefore classifies whether a high enough ρ eventually overturns the condition (Table 2). Positive entries fill the upper-left region, where the target is small, has a low import weight, or both, with an approximate trade-off between size and weight.

Table 2: Sign of q_2 under size and import-weight asymmetry

	β_{AB}						
S_B/S_C	0.10	0.20	0.35	0.50	0.65	0.80	0.90
0.1	+2.6	+3.1	+4.0	+5.4	+7.9	+15.8	+63.3
0.2	+2.7	+3.3	+4.4	+6.3	+10.2	+27.4	−
0.4	+3.0	+3.8	+5.5	+8.8	+18.9	−	−
0.7	+3.3	+4.6	+7.9	+18.2	−	−	−
1	+3.8	+5.8	+13.7	0	−	−	−
1.5	+4.8	+10.6	−	−	−	−	−
2	+6.8	+>100	−	−	−	−	−
3	−	−	−	−	−	−	−
5	−	−	−	−	−	−	−

Notes: $N = 3$, US-like parameters, $S_A = S_C = 1$, $\beta_{BA} = \beta_{CA} = 0.5$. The free-trade baseline is computed from primitives at $\bar{\rho} = 3.96$ and held fixed. "+" means $q_2 > 0$: the aggregate terms-of-trade condition eventually fails as ρ rises; the small number is the root ρ^* beyond which it fails. "−" means $q_2 < 0$, so the condition holds for all ρ in this family. "0" is the symmetric benchmark. $q_0 < 0$ and MML hold in every cell.

Takeaway. The aggregate terms of trade, and hence the first-order welfare gain from a tariff, are most likely to *deteriorate* when three forces align:

1. the target is small relative to the untariffed suppliers;
2. the target has a small weight in the tariffing country's imports;
3. foreign suppliers are close substitutes (high ρ).

In that region the bystander depreciation is both large and heavily weighted, while the gain against the target is small and lightly weighted. **A country can lower its own welfare by tariffing a small partner whose goods are easily replaced by others'.** This happens not because the tariff fails to divert expenditure, but because the diversion is so complete that the price of the replacements rises. The two-root case requires richer network asymmetry: in a representative example, $S_B/S_C = 2$, $\beta_{AB} = 0.2$, $\beta_{BA} = 0.9$ and $\beta_{CA} = 0.1$, the condition fails for $6.29 < \rho < 15.03$.

The trade-weighted NEER. The same logic applies to the NEER with trade weights. When the bystander carries most of A's trade, the NEER can depreciate under a *unilateral* tariff. Table 3 fixes $\beta_{AB} = 0.2$ and sets L_B so that w_{AC} takes the stated value. Its last column shows that the trade-weighted and import-weighted thresholds differ substantially in the same economy. In the second row, whose primitives are nearly those of example (b), the NEER depreciates only for $\rho > 3.97$, whereas the aggregate terms of trade deteriorate for $\rho > 3.40$ (3.34 at the fixed baseline of example (b)). The gap is a matter of weighting,

Table 3: Bystander reversal, NEER depreciation and aggregate terms of trade under a unilateral tariff, $N = 3$

w_{AC}	L_B	$d\nu_C > 0$ for $\rho >$	$d\nu_A^E > 0$ for $\rho >$	$R_{A\|B} > 0$ for $\rho >$
0.75	0.49	2.72	4.98	3.97
0.85	0.22	2.59	3.97	3.40
0.90	0.12	2.52	3.66	3.20

Notes: US-like parameters common to all countries, $\beta_{AB} = 0.2$, all other primitives symmetric. L_B is set so that A's baseline trade weight on C equals w_{AC} at $\rho = 3\rho_D$. Primitives are then held fixed and the free-trade baseline is re-solved at each ρ. $d\nu_A^E$ is the trade-weighted NEER; $R_{A|B}$ is the import-weighted aggregate terms of trade of Eq. (4.1). MML holds for all $\rho \leq 20$.

not of method. A buys little from B but B buys a good deal from A, so the target's weight in A's trade ($w_{AB} = 0.15$) exceeds its weight in A's imports ($b_{AB} = 0.095$). The favorable move against the target therefore counts for more in the NEER than in the welfare-relevant terms of trade.

Other casualties of asymmetry (Section C).

- *Thresholds move.* Near the symmetric point the bystander threshold rises with the target's import weight ($d\rho^*/d\beta_{AB} = +3.70$), with its size ($+0.99$) and with A's home bias ($+5.31$), and falls slightly with bystander size (-0.20), at the calibration $\alpha_D = 0.7$, $\alpha_T = 0.6$, $\eta = 1.5$ of Fig. 2; the US-like values have the same signs (Table 5).
- *Thresholds can disappear.* With primitives rather than shares held fixed, a higher ρ also shifts baseline shares toward the cheapest origin, and a large target can crowd the bystander out of A's imports. At the same calibration, with $L_B = 2.2$ the bystander reverses only for $5.99 < \rho < 53.8$, and with $L_B \geq 2.6$ it never reverses for $\rho \leq 1000$.
- *The ordering $d\nu_B < d\nu_C$ can fail*, but in every case found only with $\rho < \eta$.

4.4 Heterogeneous bystanders

With asymmetric bystanders the composite of Proposition 3.6 no longer exists. Partition the system into the target and the remaining countries R. With q denoting A's import weights within R,

$$d\nu_C^A \equiv q' d\nu_R = -q' J_{RR}^{-1} J_{RB}\, d\nu_B - q' J_{RR}^{-1} F_R\, d\tau, \tag{4.6}$$

which depends on the full internal adjustment structure of the bloc. The aggregate terms-of-trade condition still reduces to three-country form, but the bloc cannot generally be represented by a country with primitive parameters. Reversal and the failure of aggregation

share an origin. The nested structure that lets ρ direct expenditure toward other foreign suppliers is exactly what makes it matter *which* suppliers receive it. With a single, unnested elasticity the bloc aggregates exactly, but bystander reversal disappears.

4.5 A stylized reading of the 2025 episode

We do not calibrate. A static, long-run, balanced-trade model compared against monthly exchange-rate data is not a test, and Section 4.3 shows that the answer under realistic asymmetry depends on network details that a three-country stylization cannot capture. Instead we ask which way each feature of the 2025 episode pushes the trade channel. Figure 7 shows the exchange rates and the tariff timeline. Throughout, exchange-rate changes are monthly averages relative to the 2024 average, signed so that a positive number is a dollar depreciation.

Phase 1: unilateral tariffs (February–March 2025). The United States raised tariffs on China by 20 percentage points in two steps, on February 4 and March 4, without across-the-board retaliation (China's countermeasures in this phase were product-specific) and without tariffs on third countries. The configuration is a unilateral, discriminatory tariff, so average protection dominates. The model predicts dollar appreciation in effective terms, concentrated against the renminbi. In March the dollar was modestly stronger against the renminbi (-0.8 percent) and roughly unchanged against the euro (-0.1 percent).

Phase 2: retaliation (April 2025). China retaliated nearly one-for-one (US 145 percent, China 125 percent), while the United States imposed only a 10 percent baseline tariff on others. The configuration moves toward bilateral retaliation, and the bystander threshold falls from about $N\rho_D$ to ρ_D:

- *Retaliation points to dollar depreciation against bystanders.* At US-like parameters $\rho_D \approx 1.3$, below almost all cross-origin estimates. The model predicts a roughly stable USD/RMB rate together with dollar weakness against bystanders. This matches both the April pattern (the dollar 1.4 percent stronger against the renminbi but 3.8 percent weaker against the euro) and the event-study regularity of Ostry et al. (2025).
- *Expected retaliation works the same way, as a matter of interpretation.* Our model is static, so expectations enter only through the tariff structure agents expect to persist. If markets price the long-run configuration as retaliatory, the relevant comparative static is Proposition 3.5, not Proposition 3.3. In a forward-looking extension, credible

threats of retaliation would then move exchange rates before retaliation occurs. This is an interpretation, not a result of the model.

- *High cross-origin substitutability points the same way.* Trade between China and other suppliers, especially Vietnam, Mexico and other Asian producers, reallocated quickly (Fajgelbaum et al., 2024; Freund et al., 2024; Alfaro and Chor, 2023). That is the high-ρ environment in which diversion is strong.
- *China's size does not.* China is large in absolute terms and a substantial US supplier. Near the symmetric point, a larger target and a larger import weight on the target both *raise* the bystander threshold (Section 4.3). They also move the aggregate terms-of-trade condition toward its conventional sign. The configuration is therefore not clear-cut. Retaliation and ρ lean toward dollar depreciation against bystanders, while China's weight leans against it. Where the balance falls depends on bilateral shares and sizes that a three-country stylization summarizes too crudely.

Phase 3: near-uniform tariffs (late 2025). After the Geneva truce (effective May 14) and the November 10 reduction, US tariffs were about 20 percent on China and, under the framework agreements in force from August 7, 15 percent on the EU and 10–20 percent on most others. Relative treatment had largely disappeared, and the model's relevant experiment becomes average protection, which predicts mild dollar appreciation in effective terms. The dollar instead remained weak against the euro (+8.2 percent in December) and against the broad index excluding China (+2.3 percent). **The framework thus gives a reason to attribute the April pattern to trade fundamentals and refuses to attribute the December pattern to them.** Whatever drove late-2025 dollar weakness lies outside the trade channel: risk premia and tariff uncertainty, erosion of convenience yields, valuation effects, monetary policy, or expectations of renewed escalation.

Vietnam illustrates the classification point. The dollar strengthened against the dong throughout 2025 (by 3.9 percent in December), which looks inconsistent with a bystander-depreciation story. But Vietnam was itself a principal target: a 46 percent rate was announced in April, and a 20 percent rate applied from August 7, about equal to the late-2025 rate on China. With near-equal tariffs on China and Vietnam there is little relative treatment between them, so the model predicts the conventional own-tariff result.

What a quantitative general trade model would need. The analysis says what to look for in richer models. Retaliation lowers the threshold for multilateral depreciation. High cross-origin substitutability raises the diversion force. Large, heavily weighted targets

work against both. Near-uniform tariffs collapse the problem back to average protection. Quantitative multi-country models with input–output linkages (e.g., Ignatenko et al., 2025; Darracq Pariès et al., 2026) can contain all of these features. Our results identify which combinations of them determine the sign.

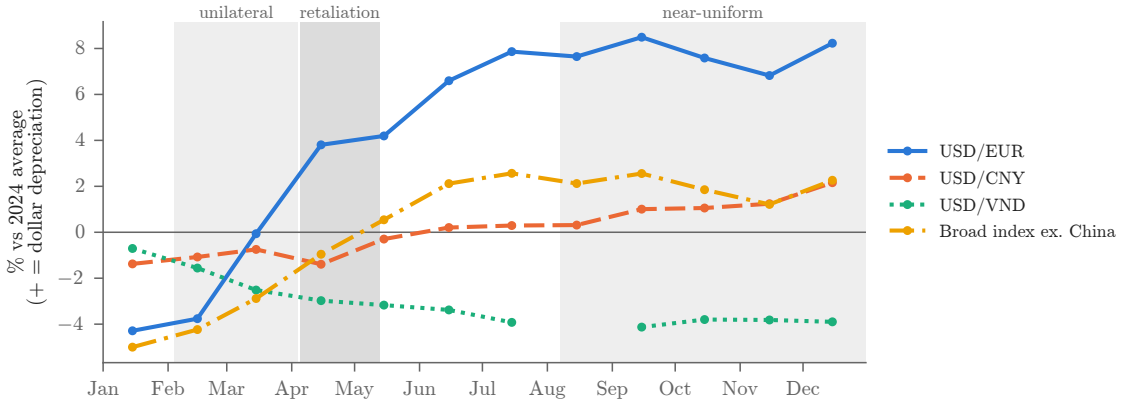
5 Conclusion

In a two-country world, the bilateral exchange rate, the effective exchange rate and the terms of trade coincide, and a tariff appreciates the tariffing country's currency wherever Marshall–Lerner holds. In a multilateral world they separate. We have characterized what the real trade channel implies once they do.

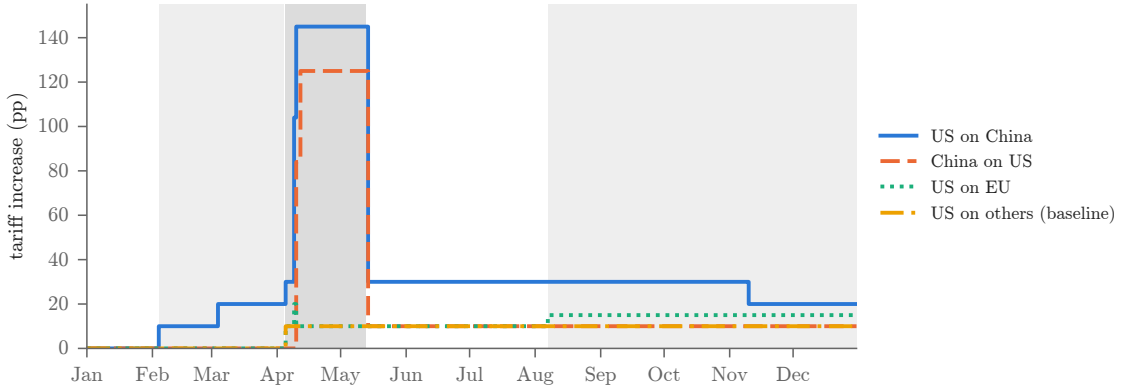
The general response is the inverse of a multilateral adjustment matrix applied to a vector of tariff disturbances. A multilateral Marshall–Lerner condition makes adjustment well behaved without fixing the sign of any bilateral response. Under symmetry, average protection moves the effective rate and relative treatment moves bilateral rates around it. In a nested-CES model the conventional result survives exactly at the effective level under unilateral tariffs. It fails bilaterally against bystanders when cross-origin substitution exceeds N times the domestic diversion margin, and against everyone under retaliation once it exceeds the domestic diversion margin itself. An exact three-country representation shows that the role of N comes from the inward-looking, highly substitutable trade of the bystander bloc, not from its size. Under asymmetry, even the effective and welfare-relevant conclusions can fail: tariffing a small, lightly weighted partner whose goods are easily replaced can worsen the tariffing country's aggregate terms of trade.

The implications run both ways. Appreciation against a tariffed partner is not evidence of broad currency strength. A multilateral depreciation during a discriminatory or retaliatory tariff episode cannot be attributed to financial channels on the basis of the two-country prediction alone. Conversely, when tariffs are nearly uniform, the trade channel predicts effective appreciation, and observed weakness must come from elsewhere.

Natural next steps are to embed the decomposition in a quantitative multi-country model with input–output linkages and heterogeneous elasticities, and to add nominal rigidities, invoicing and a financial account. This would let the long-run real benchmark derived here interact with the short-run monetary and portfolio dynamics that dominate exchange rates at business-cycle frequencies.



(a) Exchange rates, % deviation from the 2024 average (positive = dollar depreciation)



(b) Headline tariff increases above the pre-2025 baseline (percentage points)

Figure 7: The 2025 episode through the lens of the decomposition (stylized). Shading marks the unilateral phase (February 4–April 4), the retaliation phase (April 5–May 13) and the near-uniform phase (framework rates in force from August 7). (a) Monthly averages of USD per EUR, CNY and VND, and the Federal Reserve H.10 nominal broad dollar index with the renminbi removed using its published H.10 weight and renormalized (inverted so that a rise is a dollar depreciation). The August VND average is missing in the source. (b) Effective dates of headline rates: the US tariff on China, China's across-the-board tariff on the United States (China's product-specific countermeasures of February and March are not shown), the US tariff on the EU (the August rate is the 15 percent all-in rate) and the US baseline on other partners. No model magnitudes are plotted. Sources: FRED (DEXUSEU, DEXCHUS, DTWEXBGS), Federal Reserve H.10 weights, BIS (VND), Federal Register and China's State Council Tariff Commission announcements.

References

Acharya, Viral V. and Toomas Laarits, "Tariff War Shock and the Convenience Yield of US Treasuries—A Hedging Perspective," NBER Working Paper 34640, National Bureau of Economic Research 2026.

Alfaro, Laura and Davin Chor, "Global Supply Chains: The Looming 'Great Reallocation'," in "Structural Shifts in the Global Economy: Jackson Hole Economic Policy Symposium," Kansas City, MO: Federal Reserve Bank of Kansas City, 2023. Also NBER Working Paper 31661.

Arrow, Kenneth J. and Leonid Hurwicz, "On the Stability of the Competitive Equilibrium, I," *Econometrica*, 1958, *26* (4), 522–552.

_ , H. D. Block, and Leonid Hurwicz, "On the Stability of the Competitive Equilibrium, II," *Econometrica*, 1959, *27* (1), 82–109.

Artus, Jacques R. and Rudolf R. Rhomberg, "A Multilateral Exchange Rate Model," *IMF Staff Papers*, 1973, *20* (3), 591–611.

Auray, Stéphane, Michael B. Devereux, and Aurélien Eyquem, "Trade Wars, Nominal Rigidities, and Monetary Policy," *Review of Economic Studies*, 2025, *92* (4), 2228–2270.

Bagwell, Kyle and Robert W. Staiger, "An Economic Theory of GATT," *American Economic Review*, 1999, *89* (1), 215–248.

_ and _ , *The Economics of the World Trading System*, Cambridge, MA: MIT Press, 2002.

Barattieri, Alessandro, Matteo Cacciatore, and Fabio Ghironi, "Protectionism and the Business Cycle," *Journal of International Economics*, 2021, *129*, 103417.

Bergin, Paul R. and Giancarlo Corsetti, "The Macroeconomic Stabilization of Tariff Shocks: What Is the Optimal Monetary Response?," *Journal of International Economics*, 2023, *143*, 103758.

Bianchi, Javier and Louphou Coulibaly, "The Optimal Monetary Policy Response to Tariffs," NBER Working Paper 33560, National Bureau of Economic Research 2025.

Boehm, Christoph E., Andrei A. Levchenko, and Nitya Pandalai-Nayar, "The Long and Short (Run) of Trade Elasticities," *American Economic Review*, 2023, *113* (4), 861–905.

Bond, Eric W. and Constantinos Syropoulos, "The Size of Trading Blocs: Market Power and World Welfare Effects," *Journal of International Economics*, 1996, *40* (3–4), 411–437.

Broda, Christian and David E. Weinstein, "Globalization and the Gains from Variety," *Quarterly Journal of Economics*, 2006, *121* (2), 541–585.

Cardani, Roberta, Michael Kühl, Jemima Peppel-Srebrny, and Rolf Strauch, "Exchange Rate Uncertainty, Tariff Hikes, and Adjustment Costs," VoxEU.org column, 12 May 2025.

Chang, Won and L. Alan Winters, "How Regional Blocs Affect Excluded Countries: The Price Effects of MERCOSUR," *American Economic Review*, 2002, *92* (4), 889–904.

Costinot, Arnaud and Iván Werning, "Lerner Symmetry: A Modern Treatment," *American Economic Review: Insights*, 2019, *1* (1), 13–26.

Dornbusch, Rudiger, "Tariffs and Nontraded Goods," *Journal of International Economics*, 1974, *4* (2), 177–185.

Eichengreen, Barry J., "A Dynamic Model of Tariffs, Output and Employment under Flexible Exchange Rates," *Journal of International Economics*, 1981, *11* (3), 341–359.

Erceg, Christopher, Andrea Prestipino, and Andrea Raffo, "Trade Policies and Fiscal Devaluations," *American Economic Journal: Macroeconomics*, 2023, *15* (4), 104–140.

Fajgelbaum, Pablo D. and Amit K. Khandelwal, "Tariffs in 2025: Short-Run Impacts on the U.S. Economy," *Brookings Papers on Economic Activity*, 2026. Spring 2026 conference draft; also NBER Working Paper 35064.

Fajgelbaum, Pablo, Pinelopi Goldberg, Patrick Kennedy, Amit Khandelwal, and Daria Taglioni, "The US-China Trade War and Global Reallocations," *American Economic Review: Insights*, 2024, *6* (2), 295–312.

Feenstra, Robert C., Philip Luck, Maurice Obstfeld, and Katheryn N. Russ, "In Search of the Armington Elasticity," *Review of Economics and Statistics*, 2018, *100* (1), 135–150.

Freund, Caroline, Aaditya Mattoo, Alen Mulabdic, and Michele Ruta, "Is US Trade Policy Reshaping Global Supply Chains?," *Journal of International Economics*, 2024, *152*, 104011.

Gopinath, Gita, Emine Boz, Camila Casas, Federico J. Díez, Pierre-Olivier Gourinchas, and Mikkel Plagborg-Møller, "Dominant Currency Paradigm," *American Economic Review*, 2020, *110* (3), 677–719.

Hicks, John R., *Value and Capital: An Inquiry into Some Fundamental Principles of Economic Theory*, Oxford: Clarendon Press, 1939.

Ignatenko, Anna, Ahmad Lashkaripour, Luca Macedoni, and Ina Simonovska, "Making America Great Again? The Economic Impacts of Liberation Day Tariffs," *Journal of International Economics*, 2025, *157*, 104138.

Itskhoki, Oleg and Dmitry Mukhin, "The Optimal Macro Tariff," NBER Working Paper 33839, National Bureau of Economic Research 2025.

Jeanne, Olivier, "Currency Wars, Trade Wars, and Global Demand," NBER Working Paper 29603, National Bureau of Economic Research 2021.

Jiang, Zhengyang, Arvind Krishnamurthy, Hanno Lustig, Robert Richmond, and Chenzi Xu, "Dollar Upheaval: This Time Is Different," Research Paper 4241, Stanford University Graduate School of Business 2025.

Kalemli-Özcan, Şebnem, Can Soylu, and Muhammed A. Yıldırım, "Global Trade, Tariff Uncertainty, and the US Dollar," *AEA Papers and Proceedings*, 2026, *116*, 47–52. Longer version: "Tariff Uncertainty and the U.S. Dollar," NBER Working Paper 34728, 2026; also CEPR Discussion Paper 21062.

Kemp, Murray C. and Henry Y. Wan Jr., "An Elementary Proposition Concerning the Formation of Customs Unions," *Journal of International Economics*, 1976, *6* (1), 95–97.

Krugman, Paul, "The Macroeconomics of Protection with a Floating Exchange Rate," *Carnegie-Rochester Conference Series on Public Policy*, 1982, *16*, 141–182.

_ , "Is Bilateralism Bad?," in Elhanan Helpman and Assaf Razin, eds., *International Trade and Trade Policy*, Cambridge, MA: MIT Press, 1991, pp. 9–23.

Lerner, Abba P., "The Symmetry between Import and Export Taxes," *Economica*, 1936, *3* (11), 306–313.

Li, Wei, Bin Qiu, Zi Wang, Yanbo Zeng, and Yuan Zi, "Armington Elasticities and the Third-Country Effects of Trade Wars," CEPR Discussion Paper 17409, Centre for Economic Policy Research 2022.

Lindé, Jesper and Andrea Pescatori, "The Macroeconomic Effects of Trade Tariffs: Revisiting the Lerner Symmetry Result," *Journal of International Money and Finance*, 2019, *95*, 52–69.

Metzler, Lloyd A., "Stability of Multiple Markets: The Hicks Conditions," *Econometrica*, 1945, *13* (4), 277–292.

_ , "Tariffs, the Terms of Trade, and the Distribution of National Income," *Journal of Political Economy*, 1949, *57* (1), 1–29.

Mosak, Jacob L., *General-Equilibrium Theory in International Trade* number 7. In 'Cowles Commission Monograph.', Bloomington, IN: Principia Press, 1944.

Mundell, Robert A., "Flexible Exchange Rates and Employment Policy," *Canadian Journal of Economics and Political Science*, 1961, *27* (4), 509–517.

_ , "Tariff Preferences and the Terms of Trade," *Manchester School of Economic and Social Studies*, 1964, *32* (1), 1–13.

Ornelas, Emanuel, "Trade Creating Free Trade Areas and the Undermining of Multilateralism," *European Economic Review*, 2005, *49* (7), 1717–1735.

Ostry, Daniel, Simon Lloyd, and Giancarlo Corsetti, "Trading Blows: The Exchange-Rate Response to Tariffs and Retaliations," Staff Working Paper 1139, Bank of England 2025. Also CEPR Discussion Paper 20452.

Pariès, Matthieu Darracq, Aurélien Eyquem, and Valentin Jouvanceau, "The Third-Country Effects of Trade Wars," ECB Working Paper 3213, European Central Bank 2026.

Viner, Jacob, *The Customs Union Issue*, New York: Carnegie Endowment for International Peace, 1950.

Werning, Iván, Guido Lorenzoni, and Veronica Guerrieri, "Tariffs as Cost-Push Shocks: Implications for Optimal Monetary Policy," NBER Working Paper 33772, National Bureau of Economic Research 2025.

Yi, Sang-Seung, "Free-Trade Areas and Welfare: An Equilibrium Analysis," *Review of International Economics*, 2000, *8* (2), 336–347.

A The Linearized System

At a balanced free-trade baseline, let b_{ij} be origin shares within i's import bundle, h_i home shares of i's tradable spending, $y_i \equiv \tilde{I}_i$ incomes in the common currency, $s_{ij} = \alpha_T(1-h_i)b_{ij}$ the share of i's income spent on j's good, and $f_{ij} = s_{ij}\tilde{I}_i$; at the symmetric point $h_i = \alpha_D$ and $m = \alpha_T(1-\alpha_D)$. Log-differentiating the demand system, Eq. (3.5) and Eq. (3.6), with η (and, if desired, ρ) allowed to differ by country,

$$\begin{aligned}
dp_{ij} &= d\nu_j - d\nu_i + d\tau_{ij}, \qquad dp_{ii} = 0, \\
d\log P_{Mi} &= \textstyle\sum_{j\neq i} b_{ij}\, dp_{ij}, \qquad d\log P_{Ti} = (1-h_i)\, d\log P_{Mi}, \\
d\log s_{ii} &= -(1-\eta)(1-h_i)\, d\log P_{Mi}, \\
d\log s_{ij} &= (1-\rho)(dp_{ij} - d\log P_{Mi}) + (1-\eta)h_i\, d\log P_{Mi}, \\
d\log \tilde{I}_i &= d\nu_i + \textstyle\sum_{j\neq i} s_{ij}\, d\tau_{ij}, \\
d\mathrm{TB}_i &= \textstyle\sum_{k\neq i} f_{ki}(d\log s_{ki} + d\log \tilde{I}_k - d\tau_{ki}) - \sum_{j\neq i} f_{ij}(d\log s_{ij} + d\log \tilde{I}_i - d\tau_{ij}).
\end{aligned} \tag{A.1}$$

The income equation uses that at free trade $d[\tau_{ij}/(1+\tau_{ij})] = d\tau_{ij}$ and that changes in realized shares are multiplied by zero tariffs, so only the direct revenue term survives. These equations determine the matrices J and G of Eq. (2.2) at any free-trade baseline, as general functions of (y_i, h_i, b_{ij}), the elasticities and an arbitrary matrix of tariff changes. In the replication code they are implemented directly and checked against central finite differences of the nonlinear N-country solver at tariffs of $\pm 10^{-4}$, with agreement to better than 10^{-6} at symmetric and asymmetric baselines.

Lemma A.1 (Jacobian at the symmetric baseline)**.** *At the symmetric balanced baseline, where $b_{ij} = 1/(N-1)$ and $f_{ij} = m/(N-1)$, the full Jacobian is $\tilde{J}_{il} = mM_N/(N-1)$ for $l \neq i$ and $\tilde{J}_{ii} = -mM_N$. Deleting A gives*

$$-J = c\,(NI - \mathbf{1}\mathbf{1}'), \qquad c \equiv \frac{mM_N}{N-1}, \qquad j_S = c, \qquad j_D = Nc. \tag{A.2}$$

Proof. Set $d\tau = 0$, so $dp_{ij} = d\nu_j - d\nu_i$ and $d\log \tilde{I}_i = d\nu_i$. Writing $\Sigma \equiv \sum_j d\nu_j$,

$$d\log P_{Mk} = \frac{\Sigma - N\, d\nu_k}{N-1}, \qquad \frac{\partial \log P_{Mk}}{\partial \nu_l} = \frac{1 - N\delta_{kl}}{N-1}.$$

On the import side $\sum_{j\neq i}(dp_{ij} - d\log P_{Mi}) = 0$, so the $(1-\rho)$ terms cancel when $d\log s_{ij}$ is summed over $j \neq i$, leaving $\sum_{j\neq i} d\log s_{ij} = (N-1)(1-\eta)\alpha_D\, d\log P_{Mi}$. Substituting into

the trade-balance equation,

$$d\mathrm{TB}_i = \frac{m}{N-1}\Big[\sum_{k\neq i}(d\log s_{ki} + d\nu_k) - (N-1)(1-\eta)\alpha_D\, d\log P_{Mi} - (N-1)\, d\nu_i\Big].$$

Fix $l \neq i$ and differentiate term by term:

$$\frac{\partial}{\partial \nu_l}\sum_{k\neq i}\log s_{ki} = -(1-\rho)\frac{N-2}{N-1} - \frac{(1-\eta)\alpha_D}{N-1}.$$

The export income terms contribute one, the import-side price-index term contributes $-(1-\eta)\alpha_D$, and the import income term contributes zero. Collecting terms,

$$\tilde{J}_{il} = \frac{m}{(N-1)^2}\Big[(N-1) - (1-\rho)(N-2) + N\alpha_D(\eta-1)\Big] = \frac{mM_N}{N-1}.$$

The diagonal follows from homogeneity, $\tilde{J}_{ii} = -(N-1)\tilde{J}_{il} = -mM_N$. Deleting row and column A leaves diagonal entries $c(N-1)$ and off-diagonal entries $-c$, which is Eq. (A.2); its eigenvalues are c on $\mathrm{span}\{\mathbf{1}\}$ and Nc on the zero-sum subspace. □

The tariff disturbance. At unchanged exchange rates only partner i's exports to A are directly affected by a change in A's tariffs t, so

$$d\mathrm{TB}_i\Big|_{\nu} = f_{Ai}\Big(d\log s_{Ai} + d\log \tilde{I}_A - t_i\Big) = \frac{m}{N-1}\Big[-\rho_D\,\bar{t} - \rho\,(t_i - \bar{t})\Big]. \tag{A.3}$$

Indeed, $d\log P_{MA} = \bar{t}$, $d\log s_{Ai} = (1-\rho)(t_i - \bar{t}) + (1-\eta)\alpha_D\bar{t}$ and $d\log \tilde{I}_A = m\bar{t}$, and $1 - (1-\eta)\alpha_D - m = \rho_D$. Reading off the coefficients on $\bar{t}$ and $\tilde{t}_i$,

$$g_S = -\frac{m\rho_D}{N-1}, \qquad g_D = -\frac{m\rho}{N-1}, \tag{A.4}$$

and dividing by the eigenvalues in Eq. (A.2) gives $\lambda_A = -\rho_D/M_N$ and $\lambda_R = -\rho/(NM_N)$, the coefficients of Theorem 3.2.

The two-country case. At $N = 2$ each country has a single import origin, so $dp_{AB} = d\log P_{MA} = d\nu_B + d\tau$ and $dp_{BA} = d\log P_{MB} = -d\nu_B$. Then

$$d\log s_{AB} = (1-\eta)\alpha_D(d\nu_B + d\tau), \qquad d\log s_{BA} = -(1-\eta)\alpha_D\, d\nu_B,$$
$$d\log \tilde{I}_A = m\, d\tau, \qquad d\log \tilde{I}_B = d\nu_B.$$

Substituting into the trade balance, and using that the tariff wedge lowers the border value of A's imports by $d\tau$,

$$d\mathrm{TB}_A = m\Big[\big(1+2\alpha_D(\eta-1)\big)d\nu_B + \big(1-m+\alpha_D(\eta-1)\big)d\tau\Big],$$

which is the two-country system with $M_2 = 1+2\alpha_D(\eta-1)$ and $\rho_D = 1-m+\alpha_D(\eta-1) = \alpha_D\eta + (1-\alpha_D)(1-\alpha_T)$; Eq. (3.7) follows.

For the elasticity form, hold income fixed and let $M(p)$ be import volume at consumer price $p = p_{AB}$. Since import value is $s_{AB}I_A$,

$$\varepsilon_M \equiv -\frac{\partial \log M}{\partial \log p} = 1 - \frac{\partial \log s_{AB}}{\partial \log p} = 1+\alpha_D(\eta-1),$$

and by symmetry $\varepsilon_X = 1+\alpha_D(\eta-1)$. Differentiating $\mathrm{TB}_A(e,\tau) = X(1/e) - e\,M((1+\tau)e)$ at $\tau = 0$ and using balanced trade, $X = eM$, gives $\partial \mathrm{TB}_A/\partial e = M(\varepsilon_X + \varepsilon_M - 1)$. Hence $\varepsilon_X + \varepsilon_M - 1 = M_2$, and the stability condition $M_2 > 0$ is exactly $\varepsilon_X + \varepsilon_M > 1$.

The unilateral tariff. For $t = d\tau\, e_B$, $\bar{t} = d\tau/(N-1)$, $\tilde{t}_B = \frac{N-2}{N-1}d\tau$ and $\tilde{t}_C = -\frac{d\tau}{N-1}$. By Theorem 3.2,

$$d\nu_B = \frac{\lambda_A + (N-2)\lambda_R}{N-1}\,d\tau = -\frac{N\rho_D + (N-2)\rho}{N(N-1)M_N}\,d\tau,$$
$$d\nu_C = \frac{\lambda_A - \lambda_R}{N-1}\,d\tau = \frac{\rho - N\rho_D}{N(N-1)M_N}\,d\tau,$$

while $d\log e_{BC} = d\nu_C - d\nu_B = -\lambda_R\,d\tau$ and $d\nu_A^E = \lambda_A\,d\tau/(N-1)$. The bystander's disturbance follows from Eq. (A.3) with $t_C = 0$: $F_C = \frac{m}{N-1}\Big[-\frac{\rho_D}{N-1} + \frac{\rho}{N-1}\Big]$, which is Eq. (3.12).

The single-elasticity case. By the definition of ρ_D, $N\rho_D = N[1+\alpha_D(\eta-1)-\alpha_T(1-\alpha_D)]$, which rearranges to Eq. (3.11); both terms on its right-hand side are nonnegative for $\alpha_D \geq 1/N$, and equality requires $\alpha_D = 1/N$ and $\alpha_T = 1$. At $\alpha_D = 1/N$ with $\rho = \eta = \sigma$, $N\rho_D - \sigma = (N-1)(1-\alpha_T)$ and $M_N = \sigma$, so $d\nu_C/d\tau = -(1-\alpha_T)/(N\sigma)$.

Relative treatment across partners. With $\bar{t} > 0$, A depreciates against partner j if and only if

$$\frac{\rho}{N}(\bar{t} - t_j) > \rho_D\,\bar{t} \qquad \Longleftrightarrow \qquad \frac{t_j}{\bar{t}} < 1 - \frac{N\rho_D}{\rho}, \tag{A.5}$$

which follows by substituting λ_A and λ_R into $d\nu_j = \lambda_A\bar{t} + \lambda_R(t_j - \bar{t})$. The right-hand side is positive only if $\rho > N\rho_D$, so depreciation against a relatively lightly taxed partner requires both $\rho > N\rho_D$ and a sufficiently large tariff differential.

Tariffs on several partners. If A raises its tariff by $d\tau$ on a set of k partners, $1 \le k \le N-1$, then for every untargeted partner C and every targeted partner B,

$$\frac{d\log e_{AC}}{d\tau} = \frac{k(\rho - N\rho_D)}{N(N-1)M_N}, \qquad \frac{d\log e_{AB}}{d\tau} = -\frac{Nk\rho_D + (N-1-k)\rho}{N(N-1)M_N}, \qquad \frac{d\nu_A^E}{d\tau} = -\frac{k\rho_D}{(N-1)M_N}. \tag{A.6}$$

This follows by substituting $\bar{t} = k\,d\tau/(N-1)$, $\tilde{t}_B = (N-1-k)\,d\tau/(N-1)$ and $\tilde{t}_C = -k\,d\tau/(N-1)$ into Eq. (3.9). For every $k \le N-2$ the bystander response has the same threshold, $\rho > N\rho_D$.

Real exchange rates. From $q_{Aj} = e_{Aj}P_j/P_A$, with P_i the consumer price index in own currency and P_{Ni} fixed, $d\log q_{Aj} = d\nu_j + \alpha_T(1-\alpha_D)(d\log P_{Mj} - d\log P_{MA})$. For a tariff vector t imposed by A alone, symmetry gives $d\log P_{MA} = d\nu_A^E + \bar{t}$ and $d\log P_{Mj} = d\nu_A^E - \frac{N}{N-1}d\nu_j$ for a partner that imposes no tariff. Hence

$$d\log q_{Aj} = \Big(1 - \frac{mN}{N-1}\Big)d\nu_j - m\bar{t}, \qquad d\log q_{Aj} - dq_A^E = \Big(1 - \frac{mN}{N-1}\Big)\lambda_R\tilde{t}_j, \tag{A.7}$$

which with Eq. (3.9) gives Eq. (3.17). For a unilateral tariff, $d\log q_{Aj} = (1-\frac{mN}{N-1})d\nu_j - \frac{m}{N-1}d\tau$ for $j = B, C$, and averaging,

$$\frac{dq_A^E}{d\tau} = -\Big(1 - \frac{mN}{N-1}\Big)\frac{\rho_D}{(N-1)M_N} - \frac{m}{N-1}. \tag{A.8}$$

Three-country illustration. At $N = 3$, Eqs. (A.2) and (A.3) give the unilateral-tariff system explicitly:

$$J = -\frac{mM_3}{2}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \qquad F = -\frac{m}{4}\begin{pmatrix} \rho + \rho_D \\ \rho_D - \rho \end{pmatrix}, \qquad M_3 = \frac{3\alpha_D(\eta-1) + \rho + 1}{2}. \tag{A.9}$$

With $M_3 > 0$,

$$\frac{d\log e_{AB}}{d\tau} = -\frac{\rho + 3\rho_D}{6M_3} < 0, \qquad \frac{d\log e_{AC}}{d\tau} = \frac{\rho - 3\rho_D}{6M_3}, \qquad \frac{d\log e_{BC}}{d\tau} = \frac{\rho}{3M_3} > 0. \tag{A.10}$$

Thus A appreciates against B, B depreciates against C, and A depreciates against C if and only if $\rho > 3\rho_D$. By Eq. (2.4), $d\nu_C/d\tau \propto F_B + 2F_C \propto \rho - 3\rho_D$, so the sign of C's own disturbance F_C, which is positive once $\rho > \rho_D$, does not settle the sign of $d\nu_C$; Fig. 2a illustrates the intermediate case.

B Proofs

Throughout, $m \equiv \alpha_T(1-\alpha_D)$ and all derivatives are evaluated at a balanced free-trade baseline (Section A).

Proof of Lemma 2.3. Scaling every currency value by a common factor leaves all relative prices unchanged and scales every common-currency income, and hence every flow f_{ij} and trade balance TB_i, by the same factor, so TB_i is homogeneous of degree one in currency values. By Euler's theorem $\sum_l \tilde{J}_{il} = \mathrm{TB}_i$, which is zero at the balanced baseline. Summing Eq. (3.6) over i gives $\sum_i \mathrm{TB}_i \equiv 0$, so $\sum_i \tilde{J}_{il} = 0$. Under Assumption 2.1, $\tilde{J}_{il} \geq 0$ for $l \neq i$, so $\tilde{J}_{ii} = -\sum_{l\neq i} \tilde{J}_{il} \leq 0$. Deleting row and column A then gives, for each $i \neq A$,

$$|J_{ii}| - \sum_{l\neq i,A} J_{il} = \tilde{J}_{iA} \geq 0,$$

with strict inequality in every row for which $\tilde{J}_{iA} > 0$. Thus $-J$ is a Z-matrix that is weakly diagonally dominant, irreducible, and strictly diagonally dominant in at least one row. Its diagonal is strictly positive, since $|J_{ii}| = 0$ would force row i of J to vanish, contradicting irreducibility. Write $-J = D - B$ with $D = \mathrm{diag}(-J) > 0$ and $B \geq 0$. Every row sum of $D^{-1}B$ is at most one and at least one is strictly below one; with irreducibility, its spectral radius is $r < 1$. Hence $(-J)^{-1} = (I - D^{-1}B)^{-1}D^{-1} = \sum_{k\geq 0}(D^{-1}B)^k D^{-1} \geq 0$, nonsingularity follows from $r < 1$, and $\det(-J) = \det D \cdot \det(I - D^{-1}B) > 0$. □

Proof of Proposition 3.1. Let M be an $(N-1)\times(N-1)$ matrix with $PMP' = M$ for every permutation matrix P. Conjugating by the transposition of i and j gives $M_{ii} = M_{jj}$, and conjugating by a permutation that sends the ordered pair (i,j) to (k,l), $i \neq j$, $k \neq l$, gives $M_{ij} = M_{kl}$. Hence $M = aI + b\mathbf{1}\mathbf{1}'$. Since $\mathbf{1}'\mathbf{1} = N-1$ and $\mathbf{1}'v = 0$ for zero-sum v, $(aI + b\mathbf{1}\mathbf{1}')\mathbf{1} = (a + (N-1)b)\mathbf{1}$ and $(aI + b\mathbf{1}\mathbf{1}')v = av$. For $-J = a_J I + b_J \mathbf{1}\mathbf{1}'$ and $G = a_G I + b_G \mathbf{1}\mathbf{1}'$ this gives $j_S = a_J + (N-1)b_J$, $j_D = a_J$, $g_S = a_G + (N-1)b_G$ and $g_D = a_G$. For $j_S, j_D \neq 0$,

$$(-J)^{-1} = \frac{1}{j_D} I + \frac{1}{N-1}\left(\frac{1}{j_S} - \frac{1}{j_D}\right)\mathbf{1}\mathbf{1}',$$

as direct multiplication verifies. Thus $(-J)^{-1}$ and G share the common and zero-sum eigenspaces, with response coefficients $\lambda_A = g_S/j_S$ and $\lambda_R = g_D/j_D$; applying them to $t = \bar{t}\mathbf{1} + \tilde{t}$ gives Eq. (3.2). Because $\tilde{t}$ sums to zero, it does not affect the equally weighted average of the responses, which gives the first equality in Eq. (3.3); the second is immediate, and for any two partners $d\log e_{jk} = d\nu_k - d\nu_j = \lambda_R(\tilde{t}_k - \tilde{t}_j) = \lambda_R(t_k - t_j)$.

For the MML characterization, the off-diagonal entry of $-J$ is $b_J = (j_S - j_D)/(N-1)$, so $-J$ has the Z-matrix sign pattern if and only if $j_D \geq j_S$. Given $j_D \geq j_S > 0$, every entry

of $(-J)^{-1}$ above is nonnegative, so MML holds. Conversely, if $j_S \leq 0$, either $-J$ is singular or $(-J)^{-1}\mathbf{1} = \mathbf{1}/j_S$ is not nonnegative. □

Proof of Theorem 3.2. At the symmetric benchmark relabeling A's partners leaves the reduced system unchanged, so Proposition 3.1 applies. By Lemma A.1, $j_S = mM_N/(N-1)$ and $j_D = Nj_S$, so $j_S > 0$ if and only if $M_N > 0$, in which case $j_D > j_S$; by Proposition 3.1, MML is therefore equivalent to $M_N > 0$. If $\eta \geq 1$, every term in the bracket of Eq. (3.8) is nonnegative and the first is one, so $M_N > 0$. Given $M_N > 0$, dividing Eq. (A.4) by the eigenvalues in Eq. (A.2) gives $\lambda_A = -\rho_D/M_N$ and $\lambda_R = -\rho/(NM_N)$, and Eq. (3.9) is Eq. (3.2). □

Proof of Proposition 3.3. The three responses are derived in Section A. Under MML $M_N > 0$, and $\rho_D > 0$, so $d\nu_A^E < 0$ and $d\nu_B < 0$, and $d\nu_C > 0$ if and only if $\rho > N\rho_D$. Part (iv) is Eqs. (A.5) and (A.6). □

Proof of Corollary 3.4. See the single-elasticity paragraph of Section A. Since $N\rho_D \geq \eta$ when $\alpha_D \geq 1/N$, $\rho > N\rho_D$ implies $\rho > \eta$, and with $\rho = \eta$ the condition $\rho > N\rho_D$ cannot hold. □

Proof of Proposition 3.5. (i) Let primes denote responses to the mirror shock $t_{BA} = d\tau$. Relabeling $A \leftrightarrow B$ in Eq. (3.10) gives

$$d\log e'_{BA} = -\frac{N\rho_D + (N-2)\rho}{N(N-1)M_N}\,d\tau, \qquad d\log e'_{BC} = \frac{\rho - N\rho_D}{N(N-1)M_N}\,d\tau.$$

Since $e_{AB} = 1/e_{BA}$ and $e_{AC} = e_{AB}e_{BC}$, $d\log e'_{AB} = \frac{N\rho_D+(N-2)\rho}{N(N-1)M_N}d\tau$ and $d\log e'_{AC} = \frac{\rho}{NM_N}d\tau$. By linearity the combined response is $d\log e_{AB} = 0$ and

$$d\log e_{AC} = \frac{\rho - N\rho_D}{N(N-1)M_N}\,d\tau + \frac{\rho}{NM_N}\,d\tau = \frac{\rho - \rho_D}{(N-1)M_N}\,d\tau.$$

The three components are A's average protection, $-\rho_D/[(N-1)M_N]$, A's relative treatment, $\rho/[N(N-1)M_N]$, and B's relative treatment, $\rho/(NM_N)$. By symmetry between A and B the same expression applies to $d\log e_{BC}$, every bystander moves identically, so cross rates among bystanders are unchanged, and averaging over A's $N-1$ partners gives $d\nu_A^E$ in Eq. (3.13).

(ii) Superpose the N tariff vectors. A's tariff vector is uniform, $\bar{t}^{(A)} = d\tau$ and $\tilde{t}^{(A)} = 0$, and contributes $\lambda_A d\tau$ to every $d\log e_{Aj}$. Partner j taxes only A; its contribution to $d\log e_{Aj}$ is $-[\lambda_A+(N-2)\lambda_R]\frac{d\tau}{N-1}$, while each of the $N-2$ other partners k contributes $-\lambda_R d\tau = \frac{\rho}{NM_N}d\tau$,

computed as in part (i) from $e_{Aj} = e_{Ak}e_{kj}$. Summing,

$$d\log e_{Aj} = \lambda_A d\tau - \Big[\lambda_A + (N-2)\lambda_R\Big]\frac{d\tau}{N-1} - (N-2)\lambda_R d\tau = \frac{(N-2)(\rho-\rho_D)}{(N-1)M_N}\,d\tau.$$

The expression is the same for every j, so it equals the effective response, and all cross rates among A's partners are unchanged. Under MML the signs are those of $\rho - \rho_D$. □

Proof of Proposition 3.6. Label the countries A, B and bystanders $c = 1, \ldots, N-2$. Consider any tariff change in the class described in the proposition, and write the composite economy's variables with a capital C.

Step 1: bystanders move together. The baseline and the shock are invariant to permutations of the bystanders, and under MML the linear system Eq. (2.2) has a unique solution. That solution is therefore invariant to the same permutations, so $d\nu_c = d\nu_C$ for every bystander. The equilibrium is then characterized by two conditions, $d\mathrm{TB}_B = 0$ and $\sum_c d\mathrm{TB}_c = 0$ (each $d\mathrm{TB}_c$ is equal across bystanders), in the two unknowns $(d\nu_B, d\nu_C)$.

Step 2: the baseline flows coincide. In the N-country economy $f_{Ac} = f_{cA} = m/(N-1)$ for each bystander, so $\sum_c f_{Ac} = \sum_c f_{cA} = m(N-2)/(N-1)$. In the composite, $f_{AC} = \alpha_T(1-\alpha_D)\frac{N-2}{N-1} = m\frac{N-2}{N-1}$, and, since $1-h_C = 2(1-\alpha_D)/(N-1)$, $f_{CA} = \alpha_T(1-h_C)\cdot\frac{1}{2}\cdot(N-2) = m\frac{N-2}{N-1}$. The same holds for B, and $f_{AB} = f_{BA} = m/(N-1)$ in both economies. In particular the composite is balanced, and $s_{CA} = \alpha_T(1-h_C)/2 = m/(N-1) = s_{cA}$.

Step 3: A's and B's demand aggregate exactly. All bystander goods face the same price change in A, $dp_{Ac} = d\nu_C + d\tau_{Ac}$ with $d\tau_{Ac}$ common across c. Hence $d\log P_{MA} = \frac{1}{N-1}dp_{AB} + \frac{N-2}{N-1}dp_{Ac}$, which is the composite's index with $b_{AC} = \frac{N-2}{N-1}$, and $d\log s_{Ac} = (1-\rho)(dp_{Ac} - d\log P_{MA}) + (1-\eta)\alpha_D\, d\log P_{MA}$ is common across c and equal to the composite's $d\log s_{AC}$. A's income response, $d\log \tilde{I}_A = s_{AB}d\tau_{AB} + \sum_c s_{Ac}d\tau_{Ac}$, equals the composite's because $\sum_c s_{Ac} = s_{AC}$. Therefore $\sum_c df_{Ac} = df_{AC}$, and likewise df_{AB} is the same in both economies; the same argument applies to B.

Step 4: intra-bloc trade cancels. Summing Eq. (3.6) over bystanders,

$$\sum_c \mathrm{TB}_c = \sum_c (f_{Ac} + f_{Bc}) - \sum_c (f_{cA} + f_{cB}) + \Big[\sum_c \sum_{c'\neq c} f_{c'c} - \sum_c \sum_{c'\neq c} f_{cc'}\Big],$$

and the bracket is identically zero. So $\sum_c \mathrm{TB}_c$ has the form of the composite's $\mathrm{TB}_C = f_{AC} + f_{BC} - f_{CA} - f_{CB}$, and $\mathrm{TB}_B = f_{AB} + \sum_c f_{cB} - f_{BA} - \sum_c f_{Bc}$ has the form of the composite's TB_B. Given Step 3, it remains to show $\sum_c df_{cA} = df_{CA}$ and $\sum_c df_{cB} = df_{CB}$.

Step 5: the bloc's demand for A's and B's goods. Since $d\log f_{cA} = d\log s_{cA} + d\log \tilde{I}_c - d\tau_{cA}$ and all three terms are common across bystanders, $\sum_c df_{cA} = (N-2) f_{cA}\, d\log f_{cA} = f_{CA}\, d\log f_{cA}$; we need $d\log f_{cA} = d\log f_{CA}$. Incomes: $d\log \tilde{I}_c = d\nu_C + s_{cA}d\tau_{cA} + s_{cB}d\tau_{cB}$,

which equals $d\log \tilde{I}_C$ because $s_{CA} = s_{cA}$ and $s_{CB} = s_{cB}$ (Step 2). Tariff wedges: $d\tau_{cA} = d\tau_{CA}$ by construction. Shares: measure prices in bystander currency, so that other bystanders' goods have $dp = 0$, and write $dp_{cA} = \frac{S+D}{2}$ and $dp_{cB} = \frac{S-D}{2}$, where $S \equiv dp_{cA} + dp_{cB}$ and $D \equiv dp_{cA} - dp_{cB}$. A bystander's import price index moves by $S/(N-1)$, since A and B are two of its $N-1$ equally weighted origins. Hence

$$\begin{aligned} d\log s_{cA} &= (1-\rho)\Big(\frac{S+D}{2} - \frac{S}{N-1}\Big) + (1-\eta)\alpha_D \frac{S}{N-1} \\ &= (1-\rho)\frac{D}{2} + \Big[(1-\rho)\frac{N-3}{2(N-1)} + (1-\eta)\frac{\alpha_D}{N-1}\Big]S. \end{aligned}$$

In the composite the import index moves by $S/2$, so $d\log s_{CA} = (1-\rho)\frac{D}{2} + (1-\eta_C)h_C\frac{S}{2}$. The D terms coincide for any (S, D), and the S terms coincide if and only if $(1-\eta_C)(N-1)h_C = (1-\rho)(N-3) + 2\alpha_D(1-\eta)$. Using $(N-1)h_C = N-3+2\alpha_D$, this is Eq. (3.14). The argument for B's good is identical with D replaced by $-D$. The residual share on bloc goods matches as well, because shares sum to one in both economies.

Step 6: equilibrium and NEER. By Steps 2–5, the two equations $(d\mathrm{TB}_B, \sum_c d\mathrm{TB}_c) = 0$ of the N-country economy, evaluated at $d\nu_c = d\nu_C$, are identical to the composite's equations $(d\mathrm{TB}_B, d\mathrm{TB}_C) = 0$, including the tariff terms. The composite system is nonsingular (its determinant is computed below), so the two economies have the same $(d\nu_B, d\nu_C)$. The composite's trade weights are $w_{AB} = 1/(N-1)$ and $w_{AC} = (N-2)/(N-1)$, so its NEER response $\frac{1}{N-1}d\nu_B + \frac{N-2}{N-1}d\nu_C$ equals the N-country NEER at symmetric weights.

Step 7: the reduced system. By Lemma A.1, row B of $-J$ in the N-country economy is $c(N-1)$ on ν_B and $-c$ on each bystander's rate; imposing $d\nu_c = d\nu_C$ gives $(c(N-1), -c(N-2))$. Each bystander row has $-c$ on ν_B, $c(N-1)$ on its own rate and $-c$ on each of the other $N-3$ bystanders' rates; summing the $N-2$ bystander rows and imposing $d\nu_c = d\nu_C$ gives $(-c(N-2),\ (N-2)[c(N-1)-c(N-3)]) = (-c(N-2), 2c(N-2))$. This is Eq. (3.15), with determinant $c^2[2(N-1)(N-2)-(N-2)^2] = c^2N(N-2) > 0$. Inverting and applying it to $(F_B, (N-2)F_c)$ gives Eq. (3.16), and the unilateral disturbances follow from Eq. (A.3). □

The replication code verifies the reduction numerically: for $(\alpha_D, \alpha_T, \eta)$ equal to $(0.8, 0.4, 1.5)$, $(0.6, 0.7, 1.2)$, $(0.9, 0.3, 3.0)$ and $(0.5, 0.5, 2.0)$, $N \in \{4, \ldots, 10\}$ and $\rho \in \{1.1, 4, 9\}$, under the four shocks (A on B; an A–B war; A on all bystanders; all bystanders on B), the composite's $(d\nu_B, d\nu_C)$ match the N-country responses to within 10^{-12}.

Proof of Proposition 3.7. Equations Eqs. (A.7) and (A.8) are derived in Section A. For (i),

setting $d\log q_{AC} = 0$ and substituting $d\nu_C$ from Eq. (3.10) gives

$$\Big[(N-1) - mN\Big]\frac{\rho - N\rho_D}{N} = m(N-1)M_N(\rho).$$

Since M_N is affine in ρ, this is linear in ρ; for $m < 1/N$, solving gives $\rho_q^*(N)$. The coefficient $(N-1)-mN$ is then positive, and MML at ρ_q^* gives $M_N > 0$ on the right, so $\rho_q^*(N) > N\rho_D$. For (ii), under MML $M_N > 0$, and for $m < (N-1)/N$ both terms in Eq. (A.8) are negative. □

Proof of Lemma 4.1. Utility is Cobb–Douglas over the tradable aggregate and the non-tradable, so indirect utility is $V_A = \kappa\, I_A/(P_{TA}^{\alpha_T} P_{NA}^{1-\alpha_T})$ for a constant κ, with I_A, P_{TA} and P_{NA} in A's currency. The marginal utility of income is V_A/I_A, so the change in welfare in units of income is

$$\frac{dV_A}{(\partial V_A/\partial I_A)\, I_A} = d\log I_A - \alpha_T\, d\log P_{TA} - (1-\alpha_T)\, d\log P_{NA}.$$

This is the envelope theorem: by Shephard's lemma, $\alpha_T\, d\log P_{TA} = \sum_j s_{Aj}\, d\log p_{Aj}$, the expenditure-share-weighted change in consumer prices. Under the producer-price normalization $w_A = A_{TA}$ and $P_{NA} = A_{TA}/A_{NA}$ are fixed in A's currency, so $d\log P_{NA} = 0$ and the domestic tradable's price does not move. Imported goods' consumer prices move by $d\log p_{Aj} = d\nu_j + d\log(1+\tau_{Aj}) = d\nu_j + d\tau_{Aj}$ at free trade. Income is labor income plus rebated revenue, $I_A = w_A L_A/(1-\sum_j \theta_{Aj} s_{Aj})$ with $\theta_{Aj} = \tau_{Aj}/(1+\tau_{Aj})$. At free trade $\theta_{Aj} = 0$ and $d\theta_{Aj} = d\tau_{Aj}$, so changes in realized shares drop out and $d\log I_A = \sum_j s_{Aj}\, d\tau_{Aj}$: the revenue gain equals the tariff change times the border value of imports. Combining,

$$\frac{dV_A}{(\partial V_A/\partial I_A)\, I_A} = \sum_j s_{Aj}\, d\tau_{Aj} - \sum_j s_{Aj}(d\nu_j + d\tau_{Aj}) = -\sum_j s_{Aj}\, d\nu_j = -m_A \sum_{j\neq A} b_{Aj}\, d\nu_j,$$

with $m_A = \sum_j s_{Aj} = \alpha_T(1-h_A)$ and $b_{Aj} = s_{Aj}/m_A$. The tariff terms cancel: at free trade the revenue raised exactly offsets the direct price increase to first order. Tariffs imposed by other countries enter A's prices and income only through $d\nu$, so the formula holds for any tariff vector. □

The replication code checks Lemma 4.1 against central finite differences of A's log indirect utility in the nonlinear model, at symmetric and asymmetric free-trade baselines ($N = 3, 4, 5$) and under unilateral, retaliatory, multilateral-war, foreign-only and random tariff shocks; the two agree to within 10^{-6}.

Proof of Proposition 4.3. Under MML $\det A(\rho) > 0$, so $R_{A|B}$ has the sign of $Q_{A|B}(\rho)$. (a)

At the symmetric benchmark, Eq. (A.9) gives $\det A = 3(mM_3/2)^2$ and $R_{A|B} = d\nu_A^E/d\tau = -\rho_D/(2M_3)$, so $Q = -\frac{3}{8}m^2\rho_D M_3$, which is affine in ρ ($q_2 = 0$) and negative for all ρ under MML. (b) If $q_2 > 0$ and $q_0 < 0$, the product of the roots is $q_0/q_2 < 0$, so there is exactly one positive root ρ^*, with $Q < 0$ on $(0, \rho^*)$ and $Q > 0$ beyond. (c) If $q_2 < 0$, $q_1 > 0$, $q_0 < 0$ and the discriminant is positive, both roots are real with positive sum $-q_1/q_2$ and positive product q_0/q_2, so both are positive, and the downward parabola is positive only between them. □

C Away from the Symmetric Free-Trade Benchmark

The results of Section 3 are local to the symmetric free-trade benchmark. This appendix examines four departures from it: a positive common external tariff, finite tariff changes, asymmetric primitives, and the trade-weighted effective rate. The representation Eq. (2.3) and the MML condition apply at any balanced baseline. What changes is the mapping from primitives to the eigenvalues of Proposition 3.1 and, once partner symmetry fails, whether the common and zero-sum directions remain eigen-directions at all. In every case the baseline equilibrium is recomputed from primitives rather than by perturbing shares, and MML holds at every point reported. Unless stated otherwise, numerical values use the calibration of Fig. 2, $\alpha_D = 0.7$, $\alpha_T = 0.6$ and $\eta = 1.5$ ($\rho_D = 1.17$); US-like values are reported where they are discussed in the main text.

C.1 A positive common external tariff

Suppose A levies a common tariff τ_0 on all $N-1$ partners, which impose no tariffs of their own. This baseline treats the partners alike, so partner symmetry holds and Proposition 3.1 applies. Relative to free trade, A's currency appreciates by the same amount against every partner, so A's goods are more expensive abroad and baseline shares differ from their free-trade values. We measure tariff changes as $t_j = d\log(1+\tau_{Aj})$, which coincides with $t_j = d\tau_{Aj}$ at free trade. Let $\theta \equiv \tau_0/(1+\tau_0)$ denote the tariff wedge, μ_A the import share of A's tradable expenditure, μ_P the import share of a partner's tradable expenditure, and ω the share of A's good in a partner's import bundle. The rebate channel enters through $\phi \equiv \alpha_T\mu_A/(1-\theta\alpha_T\mu_A)$, the response of A's income to its import expenditure. Trade between A and each partner is balanced at the baseline, so $X \equiv f_{Aj} = f_{jA}$ denotes this bilateral flow, and $Y \equiv f_{jk}$ the flow between any two partners. Linearizing at this baseline gives

$$g_S = -X\rho_D(\tau_0), \qquad g_D = -X\rho, \qquad j_S = XM_S, \qquad j_D = \Big(X + (N-1)Y\Big)M_D, \tag{C.1}$$

where

$$\begin{aligned}\rho_D(\tau_0) &= 1-\phi(1-\theta)+(\eta-1)(1-\mu_A)(1+\phi\theta),\\ M_S &= 1+(\eta-1)(1-\mu_A)(1+\phi\theta)+(\rho-1)(1-\omega)+(\eta-1)(1-\mu_P)\,\omega,\\ M_D &= 1+(\rho-1)\frac{N-3+\omega}{N-2}+(\eta-1)(1-\mu_P)\frac{N-1-\omega}{N-2}.\end{aligned}$$

At free trade $\theta = 0$, $\phi = m$, $\mu_A = \mu_P = 1-\alpha_D$ and $\omega = 1/(N-1)$, so $\rho_D(0) = \rho_D$ and $M_S = M_D = M_N$.

The eigenvalues in Eq. (C.1) differ in the trade flows on which they act. Tariff disturbances along both components fall only on partners' trade with A, so g_S and g_D are proportional to X. A common exchange-rate movement leaves cross rates among partners unchanged and also shifts only partners' trade with A, so j_S is proportional to X, and X cancels from $\lambda_A = -\rho_D(\tau_0)/M_S$. A relative movement also shifts trade among partners. Under a zero-sum movement v, partner j's rate moves by v_j against A and by $v_j - v_k$ against partner k; since $\sum_{k\neq j} v_k = -v_j$, its average movement against the other partners is $(N-1)/(N-2)$ times v_j. The base $X+(N-1)Y$ is therefore partner j's trade with A plus its trade with the other partners, the latter weighted by this factor. Define $x \equiv X/(X+(N-1)Y)$, A's share of the trade on which relative exchange-rate movements act. Then $\lambda_R = -x\rho/M_D$ and $j_D/j_S = (1/x)(M_D/M_S)$. At free trade $X = Y$, so $x = 1/N$ and $j_D = Nj_S$: the factor N in the bystander threshold is $1/x$ evaluated at the free-trade baseline.

For a further tariff on B, $t = dt\, e_B$, the bystander response is $d\nu_C = (\lambda_A - \lambda_R)\, dt/(N-1)$, so A depreciates against C if and only if

$$\rho > \rho^*(\tau_0) \equiv \frac{\rho_D(\tau_0)}{x}\,\frac{M_D}{M_S}. \tag{C.2}$$

The condition is implicit in ρ, because ω, μ_P, x and M_S depend on baseline shares that themselves depend on ρ; we compute it by root-finding with the baseline re-solved at each ρ. Since the response is linear in t, a marginal preferential cut for B, $t = -dt\, e_B$, worsens C's terms of trade under the same condition. Table 4 reports the threshold relative to its free-trade value.

The threshold rises with τ_0, mainly because x falls. The common appreciation makes A's goods more expensive, so A's share of each partner's imports falls. At $N = 3$ and $\tau_0 = 1$, ω falls from $1/2$ to 0.29 and x from $1/3$ to 0.17. Relative treatment then disturbs a smaller part of the trade on which relative exchange-rate movements act. A rise in M_S relative to M_D offsets part of this effect. The local threshold also understates what a discrete preferential cut requires. In the exact model at $N = 3$, removing the tariff on B from $\tau_0 = 0.5$ worsens

Table 4: Bystander threshold at a common external tariff, $\rho^*(\tau_0)/N\rho_D$

	$\tau_0 = 0.25$	$\tau_0 = 0.5$	$\tau_0 = 1$
$N = 3$	1.12	1.22	1.41
$N = 5$	1.23	1.49	2.07

Notes: $\alpha_D = 0.7$, $\alpha_T = 0.6$, $\eta = 1.5$. Computed from Eq. (C.2) with the baseline re-solved at each ρ. At US-like parameters the ratios are 1.10, 1.19, 1.36 ($N = 3$) and 1.23, 1.48, 2.07 ($N = 5$).

C's terms of trade only if $\rho > 6.23$, compared with 4.28 for a marginal cut at the same baseline and $3\rho_D = 3.51$ at free trade.

C.2 Finite tariffs

The thresholds of Propositions 3.3 and 3.5 concern marginal tariffs at free trade. In one case the finite-tariff sign is exact. At $N = 2$ and $\eta = 1$, expenditure shares are fixed at their weights, and $I_A = w_A L_A/[1 - \frac{\tau}{1+\tau}m]$, so $\mathrm{TB}_A = 0$ becomes $e\, m\, w_B L_B = m I_A/(1+\tau)$:

$$e(\tau) = \frac{w_A L_A}{w_B L_B} \cdot \frac{1}{1 + (1-m)\tau}, \qquad \left.\frac{d \log e}{d\tau}\right|_{\tau=0} = -(1-m) = -\left.\frac{\rho_D}{M_2}\right|_{\eta=1}.$$

Thus $e(\tau)$ is decreasing in τ, and the sign of the linearized response is exact for all tariff levels in this case.

For finite tariffs at $N = 3$, starting from the symmetric free-trade benchmark, let $\rho^*(\tau)$ denote the value of ρ at which $\log e_{AC}(\tau) = 0$ in the exact model. Writing $\log e_{AC}(\tau) = a(\rho)\tau + \frac{1}{2} b_C \tau^2 + O(\tau^3)$, where a is the first-order slope and b_C the second derivative along the tariff path,

$$\rho^*(\tau) = \rho_0^* + \kappa\, \tau + O(\tau^2), \qquad \kappa = -\frac{b_C}{2a'(\rho_0^*)}.$$

For a unilateral tariff, $a = (\rho - 3\rho_D)/(6M_3)$ by Eq. (A.10), so $\kappa_{\text{uni}} = -3M_3 b_C$. For bilateral retaliation, $\tau_{AB} = \tau_{BA} = \tau$ and $a = (\rho - \rho_D)/(2M_3)$ by Eq. (3.13), so $\kappa_{\text{war}} = -M_3 b_C$. Both are evaluated at the symmetric benchmark, where b_C follows from a second-order expansion of the equilibrium conditions.

The unilateral threshold falls with the size of the tariff. On a grid over $\alpha_D \in [1/3, 1)$, $\alpha_T \in (0, 1]$ and $\eta \in [1, 10]$, the exact finite-difference slope $(\rho^*(0.01) - 3\rho_D)/0.01$ is negative at every point. At $\tau = 1$ the exact threshold is 0.76 times $3\rho_D$ at the calibration of this appendix and 0.72 times $3\rho_D$ at US-like parameters; at the 2025 peak rate $\tau = 1.45$ it is 2.70 at US-like parameters, against $3\rho_D = 3.96$. A larger tariff shrinks B's share of A's imports, so a further increase adds less average protection while continuing to divert expenditure

toward C. The retaliation threshold instead rises slightly, with

$$\kappa_{\text{war}} = \frac{1-\alpha_D}{4}\Big[\alpha_D(\eta-1)^2 + 3\alpha_D\alpha_T(\eta-1) + 2\alpha_T(1-m)\Big] > 0 \qquad (\eta \geq 1),$$

and at $\tau = 1$ the exact threshold is 1.07 times ρ_D (1.03 at US-like parameters).

These two subsections illustrate a common principle. The threshold depends on the share structure on which the tariff acts. A common tariff shrinks A's footprint in its partners' trade and raises the threshold, while a concentrated tariff shrinks the target's share of A's imports and lowers it.

C.3 Asymmetric primitives

At a general balanced baseline the coefficients of Eq. (A.1) are the baseline shares b_{ij}, the home shares h_i and the incomes y_i. The elasticity ρ enters the share responses only through $(1-\rho)$, so every entry of J and F is affine in ρ. Each entry of $\text{adj}(-J)$ is therefore a polynomial of degree at most $N-2$ in ρ, and each numerator $[\text{adj}(-J)F]_i$ a polynomial of degree at most $N-1$. Since $\det(-J) > 0$ under MML, a bilateral response can change sign only at a root of this numerator. Symmetry reduces the problem to the two coefficients of Proposition 3.1, yielding linear thresholds. At a general asymmetric baseline the $N = 3$ numerator is already quadratic in ρ:

$$\text{sign}\,\frac{d\log e_{AC}}{d\tau} = \text{sign}\left(c_2\rho^2 + c_1\rho + c_0\right), \qquad c_2 = \alpha_T^2\, b_{AB}b_{AC}b_{CA}b_{CB}(1-h_A)(1-h_C)y_A y_C. \tag{C.3}$$

The coefficients c_1 and c_0 depend on the full pattern of trade shares and country sizes. Provided MML holds, $c_2 > 0$ whenever A and C each import from both foreign origins, so at given shares the response is positive beyond the largest positive root, which equals $3\rho_D$ at the symmetric point.

The quadratic holds at given shares, however, and the shares themselves depend on ρ once primitives are held fixed. At an asymmetric baseline relative prices differ across origins, and a higher ρ also reallocates expenditure toward the cheapest origin. A large target, whose currency must depreciate to balance its trade, can then crowd C out of A's imports and prevent the reversal. With $L_B = 2.2$ and all other primitives symmetric, e_{AC} reverses only for $5.99 < \rho < 53.8$, as A's import share from B rises from 0.74 at $\rho = 5$ to 0.997 at $\rho = 100$; with $L_B \geq 2.6$ it does not reverse for any $\rho \leq 1000$. Away from symmetry there may therefore be no bilateral threshold, one, or several, and for $N \geq 4$ the numerator is of higher degree in ρ.

The threshold also moves with asymmetric primitives near the symmetric benchmark.

Table 5 reports $d\rho^*/dx$ at $\rho = 3\rho_D$, where ρ^* solves $d \log e_{AC}/d\tau = 0$ after perturbing a single primitive x and recomputing the free-trade baseline. Derivatives are evaluated at $L_i = 1$ and $\beta_{AB} = 1/2$. Here β_{AB} is A's weight on B, with $\beta_{AC} = 1 - \beta_{AB}$, and α_D^A is the home weight in A alone.

Table 5: Sensitivity of the bystander threshold to asymmetric primitives, $N = 3$

	$d\rho^*/dx$	
x	$(\alpha_D, \alpha_T, \eta) = (0.7, 0.6, 1.5)$	US-like $(0.8, 0.4, 1.5)$
β_{AB}	+3.70	+4.06
L_B	+0.99	+1.23
L_C	−0.20	−0.32
α_D^A	+5.31	+6.61

The interpretations are as follows. β_{AB}*:* a tariff on a larger supplier is a larger average-protection shock, so B's loss rises relative to C's gain. L_B*:* a larger target depreciates to balance its trade, which raises A's import share from B. L_C*:* a larger bystander depreciates and gains import share in A, but also absorbs its own disturbance more easily; the two effects nearly offset. α_D^A*:* stronger home bias in A raises domestic diversion and makes C's rate respond more to B's loss.

C.4 The effective rate and the ordering of bilateral responses

At the symmetric benchmark, relative treatment has no effect on the equally weighted average of A's bilateral responses, and this average coincides with the trade-weighted NEER. A unilateral tariff therefore always appreciates A's effective exchange rate. Relative treatment drops out of the effective rate only when the trade weights and the adjustment structure are symmetric. With asymmetric trade weights, $d\nu_A^E = w_{AB}\, d\nu_B + w_{AC}\, d\nu_C$, and a bystander depreciation $d\nu_C > 0$ enters with weight w_{AC}. The trade-weighted NEER can then depreciate when the bystander reversal occurs and the bystander carries most of A's trade weight. Table 3 illustrates this at US-like parameters. The reversal starts below $3\rho_D$ because a small target with a small import weight lowers the threshold, as in Table 5. The exact model at $\tau = 0.01$ confirms the signs.

The ordering $d\nu_B < d\nu_C$, which holds at the symmetric benchmark because $\lambda_R < 0$, can also fail. This requires ρ near one and a high η, and $\rho < \eta$ in every case we examined. For example, with $L = (5.12, 4.26, 2.28)$ for (A, B, C), $\beta_{AB} = 0.86$, $\beta_{BA} = 0.65$, $\beta_{CA} = 0.19$, $\alpha_D = 0.89$, $\alpha_T = 0.6$, $\eta = 3.74$ and $\rho = 1.20$, $d\nu_B - d\nu_C = 0.08\, d\tau$ (MML holds; the exact model at $\tau = 0.01$ gives 0.079).

Online Appendix

D Further Robustness

D.1 Robustness of the two-country result

This subsection collects the monotonicity results behind Remark 2.1. Equilibrium is balanced trade for A, $\mathrm{TB}_A(e,\tau) \equiv X_A(e,\tau) - M_A(e,\tau) = 0$, where e is units of A's currency per unit of B's, τ is A's ad valorem tariff on imports from B, and X_A and M_A are the values of exports and imports in A's currency. Assume TB_A is continuously differentiable with $\mathrm{TB}_{A,e} \neq 0$ locally. By the implicit function theorem there is a locally unique mapping $e(\tau)$ with

$$\frac{de}{d\tau} = -\frac{\mathrm{TB}_{A,\tau}(e(\tau),\tau)}{\mathrm{TB}_{A,e}(e(\tau),\tau)}.$$

Under the producer-price normalization the consumer price of imports in A is $p_M(e,\tau) = (1+\tau)e$ and the price of A's exports in B is $p_X(e) = 1/e$.

Lemma D.1 (Tariff monotonicity)**.** *Fix e. Suppose (i) $p_M(e,\tau)$ is increasing in τ, and (ii) import expenditure $M_A(e,\tau)$ is decreasing in the consumer import price. Suppose in addition that any induced effect of τ on export revenue satisfies $X_{A,\tau} \geq 0$, or the weaker condition $X_{A,\tau} > M_{A,\tau}$. Then $\mathrm{TB}_{A,\tau} > 0$.*

Proof. $\mathrm{TB}_{A,\tau} = X_{A,\tau} - M_{A,\tau}$, and by (i)–(ii) $M_{A,\tau} = (\partial M_A/\partial p_M)(\partial p_M/\partial \tau) < 0$. Either condition on $X_{A,\tau}$ then gives $\mathrm{TB}_{A,\tau} > 0$. □

In the baseline model the tariff applies only to imports and there is no policy response abroad, so $X_{A,\tau} = 0$ at fixed e and the condition holds trivially.

Lemma D.2 (Exchange-rate monotonicity)**.** *Fix τ. If $M_{A,e} < 0$ and $X_{A,e} \geq 0$, with at least one inequality strict, then $\mathrm{TB}_{A,e} > 0$.*

Proof. $\mathrm{TB}_{A,e} = X_{A,e} - M_{A,e} > 0$ under the stated conditions. □

Proposition D.3 (Sign of the two-country response)**.** *If the conditions of Lemmas D.1 and D.2 hold in a neighborhood of an equilibrium (e^*,τ), then $e(\tau)$ exists locally and $de/d\tau < 0$.*

Proof. By the implicit function theorem and Lemmas D.1 and D.2, $de/d\tau = -\mathrm{TB}_{A,\tau}/\mathrm{TB}_{A,e} < 0$. □

The condition of Lemma D.2 is the Marshall–Lerner condition, and that of Lemma D.1 is its tariff analogue: import expenditure falls as the consumer import price rises. Both hold across standard specifications.

Demand. Tariffs and exchange rates affect demand only through relative prices. Under CES (Armington) preferences over tradables with any elasticity $\sigma > 0$, separable from non-tradables, Marshallian import demand is strictly decreasing in p_M, so at fixed e a higher tariff lowers import expenditure while export revenue is locally unaffected; holding τ fixed, a depreciation raises the domestic-currency price of imports and lowers the foreign-currency price of exports. The same holds under nested CES with separate elasticities between tradables and non-tradables and across tradable varieties: tariff-induced income effects may reallocate expenditure between tradables and non-tradables, but the monotone relation between import demand and its consumer price is preserved. Under quasi-linear preferences income effects from tariff revenue vanish and the conditions hold by construction.

Production. With diminishing returns or sector-specific factors, wages and output adjust to demand. At fixed e a higher tariff still raises the consumer price of imports and reduces import expenditure, and since the tariff introduces no export-side wedge, export revenue is not adversely affected. Holding the tariff fixed, a depreciation raises import prices and lowers export prices; provided export supply is weakly increasing in the export price, import expenditure falls and export revenue weakly rises, so $\mathrm{TB}_{A,e} > 0$ continues to hold locally.

Fiscal closure and pricing. Whether tariff revenue is rebated lump-sum, wasted, transferred abroad or used to finance government spending, a tariff raises the consumer price of imports at a fixed exchange rate, so the conditions of Lemmas D.1 and D.2 hold under standard fiscal closures. Local- or dominant-currency pricing and monopolistic competition mainly affect the degree and timing of pass-through; in the flexible-price equilibrium that our analysis describes, consumer prices reflect both the tariff wedge and the exchange rate, and the monotonicity conditions remain valid. While the magnitude of the response varies across specifications, its sign does not, wherever Marshall–Lerner holds.

D.2 Sector-specific factors

The linear technology makes export supply infinitely elastic at the going wage. To formalize the supply side, replace $Y_{Ti} = A_{Ti}L_{Ti}$ with $Y_{Ti} = A_{Ti}L_{Ti}^{\gamma}$, $\gamma \leq 1$, as in the Ricardo–Viner model, with non-tradables linear as before and labor mobile across sectors within each country. Competitive pricing gives $w_i = \gamma A_{Ti}L_{Ti}^{\gamma-1}$ under the producer-price normalization,

and the fixed factor earns rents $(1-\gamma)Y_{Ti}$ that are distributed lump-sum to households. The sectoral labor allocation now responds to the tariff, and $\gamma = 1$ recovers the baseline exactly.

Upward-sloping export supply loads diverted demand onto the bystander's price, which under the producer-price normalization is its currency. Diminishing returns therefore lower the bystander threshold rather than raise it. At the symmetric point the effect is modest: the threshold falls from 3.96 to [TODO: 3.83 (earlier draft; not recomputed with the current code)] at $\gamma = 0.5$. It is larger for small bystanders, which cannot expand export quantities cheaply: for a bystander of Vietnam's relative size the threshold falls from [TODO: 8.76] to [TODO: 5.27] at $\gamma = 2/3$ (earlier draft, under that draft's calibration; not recomputed). None of the sign results of the paper requires diminishing returns; the linear baseline delivers them.

D.3 Permanent trade imbalances

Balanced trade is a modelling choice, not a requirement of the representation Eq. (2.3), which applies at any baseline. With a permanent transfer that finances a trade deficit for A, the threshold structure is unchanged but its location moves with the baseline. A US-scale deficit of 3 percent of income raises the symmetric-point bystander threshold from 3.96 to [TODO: 4.6–5.1, depending on whether the counterpart surplus is B's or C's (earlier computations; not recomputed)]. At 2025 tariff levels, however, the effect of tariff size dominates: a finite unilateral tariff of $\tau = 1.45$ lowers the threshold to 2.70 at a balanced baseline (Section C.2), and the earlier computations put the combined real-rate, deficit-adjusted, discrete-tariff threshold at [TODO: about 3.5 (3.55 for a 3% deficit at $\tau = 1.45$; not recomputed)].